

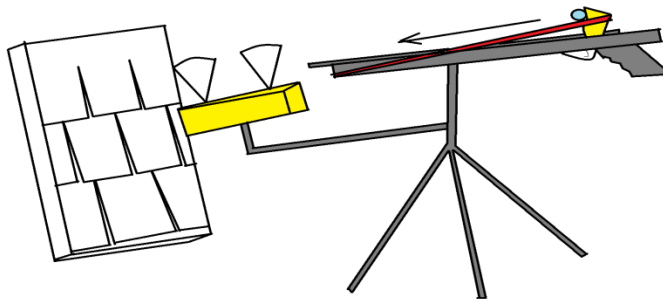
A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

Ice ball impact laboratory test research has claimed to show the potential damage causing capability for hail stones of various sizes by correlating test results that have recently been conducted to simulate hail stone impacts at perpendicular striking angles with different types of selected roofing materials. These researchers chose to test 9 different spheres of ice at 9 different targeted "Free Fall Velocities" to find what size ice balls were capable of producing damage to various types of selected roofing materials.

What do Ice ball tests that takes place in a laboratory, and a real life hail storm have in common?

How are hail storm scenarios re-created in a laboratory?

- 1) Perfectly spherical ice balls are created in a laboratory
- 2) Predetermined sizes are selected
- 3) The ice balls are propelled Horizontally at roofing materials
- 4) Desired "Free Fall" speeds increase with the size of the ice spheres
- 5) The Ice Sphere's pass threw a speed measuring chronograph
- 6) The Ice Spheres impact the selected roofing materials at Perpendicular impacts angles.
- 7) The mass of the ice spheres and the speed at which they are being propelled is then used to determine the amount of kinetic energy that is necessary to inflict damage
- 8) Results are then correlated to Real life scenarios



Do hail stones impact roofs this way during a severe thunderstorm?

What would a hail bearing storm look like if it produced hail stones that were made out of perfectly spherical ice balls of pre selected sizes, that fell at specific and targeted free fall speeds that would produce perpendicular impacts on every slope of an actual roof?

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The following sizes and targeted “Free Fall Velocities” were selected for the testing.

1” diameter ice balls were propelled at a desired “Free Fall Velocity” of 50 miles per hour.

1.25” diameter ice balls were propelled at a desired “Free Fall Velocity” of 56 miles per hour

1.5” diameter ice balls were propelled at a desired “Free Fall Velocity” of 61 miles per hour

1.75” diameter ice balls were propelled at a desired “Free Fall Velocity” of 66 miles per hour

2” diameter ice balls were propelled at a desired “Free Fall Velocity” of 72 miles per hour

2.25” diameter ice balls were propelled at a desired “Free Fall Velocity” of 76 miles per hour

2.5” diameter ice balls were propelled at a desired “Free Fall Velocity” of 80 miles per hour

2.75” diameter ice balls were propelled at a desired “Free Fall Velocity” of 84 miles per hour

3” diameter ice balls were propelled at a desired “Free Fall Velocity” of 88 miles per hour

By definition, **free falling objects** are considered to be objects that are falling **through a median of negligible resistance, relative to the sole influence of gravity alone**. An example of this would be an object which is falling inside of a vacuum tunnel where no external resistance or acceleration will be allowed to influence the acceleration of the falling object such as air density, drag, viscosity, or external sources of wind. **Gravity is the only factor** that will set a free falling object into motion. On earth, objects that fall through a median of negligible resistance due to the sole influence of gravity will fall at the rate of approximately 9.81 meters per second squared. Therefore, to find the comparable heights that would be necessary to achieve these 9 desired free fall velocities that these researches elected to test with, calculations were made to see how high free falling objects would need to be dropped from in order to achieve these 9 desired free fall velocities, in order to determine how comparable these heights would compare to altitudes that are commonly associated with actual hail storms. On average, hail storms typically occur between altitudes of 40,000 to 70,000 feet high.

To achieve this the 50 miles per hour desired free fall target speed for a 1 inch diameter ice ball, was converted to feet per second which is 73.33333333 FPS, and then divided by the rate of gravity that was also converted to 32.1850393 FPS to figure out how long it would take an object to fall before reaching 50 miles per hour, which is 2.27849 seconds, and then the square of the fall time (2.27849^2) which is 5.19152 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 83.544 feet.

The 56 miles per hour desired free fall target speed for a 1.25 inch diameter ice ball, was converted to feet per second which is 82.13333333 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 56 miles per hour, which is 2.5519103 seconds, and then the square of the fall time was taken (2.5519103^2) which is 6.51224618, which was then multiplied this by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 104.798 feet.

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The 61 miles per hour desired free fall target speed for a 1.5 inch diameter ice ball, was also converted to feet per second which is 89.4666666666667 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 61 miles per hour, which is 2.77975434 seconds, and then the square of the fall time (2.77975434^2) which is 7.72706251 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 124.347 feet.

The 66 miles per hour desired free fall target speed for a 1.75 inch diameter ice ball, was converted to feet per second which is 96.8 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 66 miles per hour, which is 3.07760856 seconds, and then the square of the fall time (3.07760856^2) which is 9.0457093 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 145.56 feet.

The 72 miles per hour desired free fall target speed for a 2 inch diameter ice ball, was converted to feet per second which is 105.6 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 72 miles per hour, which is 3.28102753 seconds, and then the square of the fall time (3.28102753^2) which is 10.765414165 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 173.238 feet.

The 76 miles per hour desired free fall target speed for a 2.25 inch diameter ice ball, was converted to feet per second which is 111.46666 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 76 miles per hour, which is 3.46330683 seconds, and then the square of the fall time (3.46330683^2) which is 11.99449424 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 193.021 feet.

The 80 miles per hour desired free fall target speed for a 2.5 inch diameter ice ball, was converted to feet per second which is 117.3333333 FPS, and then divided by the rate of gravity that was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 80 miles per hour, which is 3.64558614 seconds, and then the square of the fall time (3.64558614^2) which is 13.29029833 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 213.874 feet.

The 84 miles per hour desired free fall target speed for a 2.75 inch diameter ice ball, was converted to feet per second which is 123.2 FPS, and then divided by the rate of gravity was also converted to FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 84 miles per hour, which is 3.82786545 seconds, and then the square of the fall time (3.82786545^2) which is 14.65255391 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 235.796 feet.

The 88 miles per hour desired free fall target speed for a 3 inch diameter ice ball, was converted to feet per second which is 129.066666 FPS, and then divided by the rate of gravity that was also converted to

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FPS of 32.1850393 to figure out how long it would take an object to fall before reaching 88 miles per hour, which is 4.01014475 seconds, and then the square of the fall time (4.01014475^2) which is 16.08126098 was multiplied by half the rate of gravity converted to feet per second 16.0925 which equals a drop height of 258.788 feet.

The results showed that the velocities that were being tested with, were equivalent with what would occur if each of the ice balls were falling from within a vacuum tunnel from heights that would be between **84 to 258 feet high**, without any influence from wind or any other resistance including drag, and without any consideration for differences in air density or changes in the viscosity of the objects during their fall towards the earth. These researchers also elected to test with ice balls that were perfectly spherical, despite the fact that real life hail stones are not perfectly spherical and are often found to be random in shape. During the tests, each ice ball impacted the materials that were being tested with at perpendicular striking angles, despite the fact that it would be impossible for an object to free fall vertically while striking a sloped roof at a perpendicular striking angle. **What would a hail storm look like if it were to occur similarly to how these researchers have been conducting their impact tests during laboratory research?**

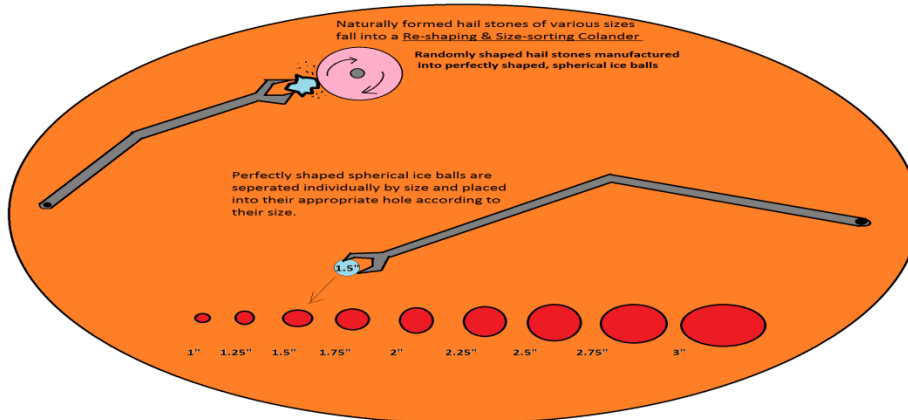


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Let's try to replicate what a hail storm would look like if it were to produce hail stones that were perfectly spherical, made only of specific sizes, that always fell at specific "Free Fall Speeds," and that would strike the roof slopes at perpendicular impact angles.

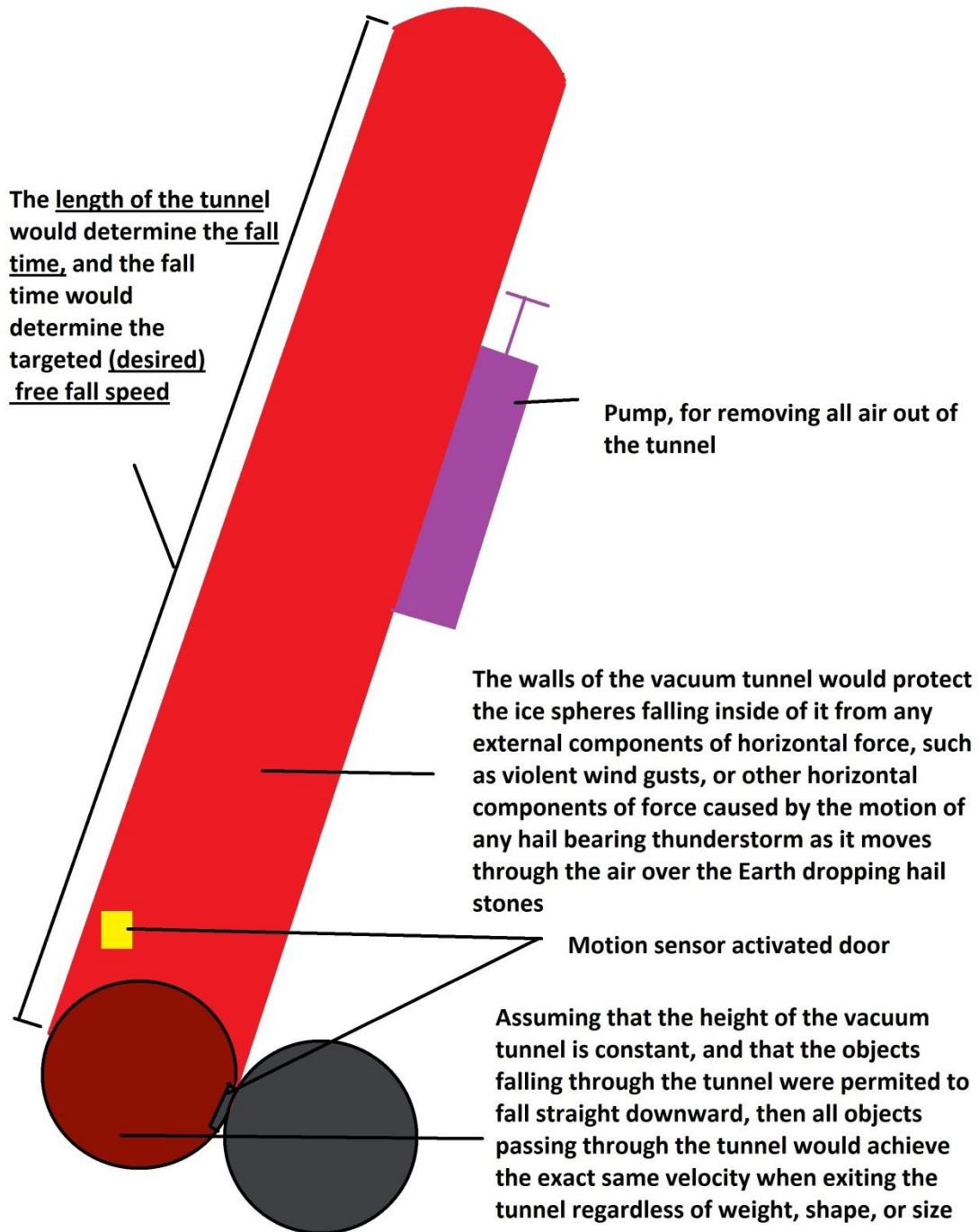
I suppose that the first thing that would be needed in order to achieve this would be for the hail storm to drop the hail stones into a large re-shaping and sorting collander that would have to exist above the roof some how that could re-manufacture the randomly shaped hail stones into perfect spheres, and that could also sort the perfectly shaped spheres according to their specific sizes, so that the ice sphere of a specific size could then be placed into a vacuum tunnel of the specific length that would be necessary to enable each ice sphere of a selected size to achieve the specific "Desired Free Fall Speed" being targeted before striking the roof at a perpendicular angle of impact.



This remanufacturing and sorting machine, would need to be able to make sure that the 1 inch diameter ice sphere's always get placed into the 1 inch diameter vacuum tunnel tube, and that the 2.25 inch diameter ice spheres always get placed into the 2.25 inch diameter vacuum tunnel tube, in order to make sure that the exact free fall velocity is achieved for each perfectly spherical hail stone according to its size. For example, it would screw everything up if the 1.25 inch diameter ice sphere ended up in the 3 inch vacuum tunnel because it would strike the roof at 88 mph instead of 56 mph.

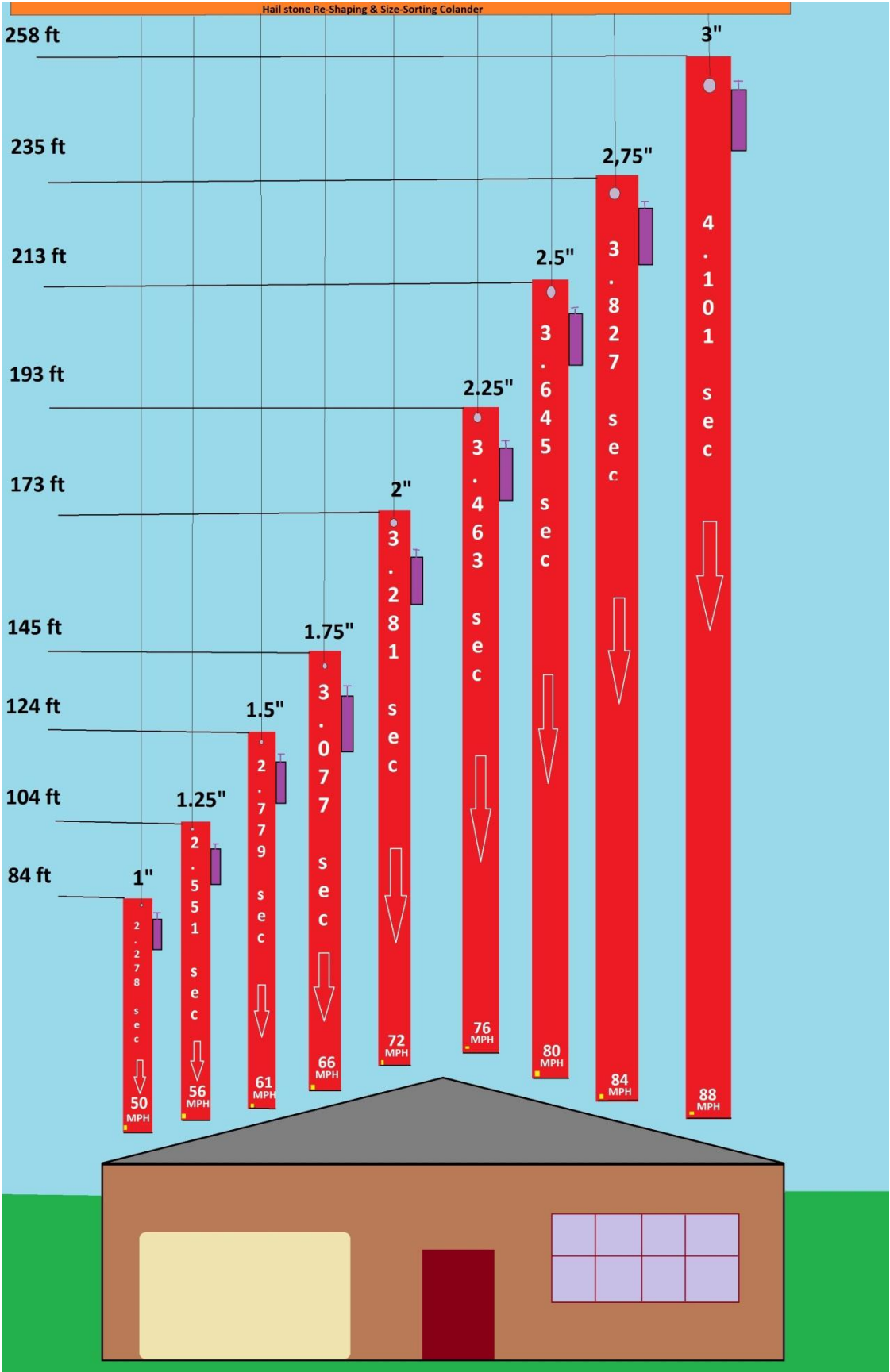
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Then we would need 9 different vacuum tunnels of the correct lengths to drop each hail stone into that would enable the hail stones to reach their targeted Free Fall Speeds for each of their specific sizes



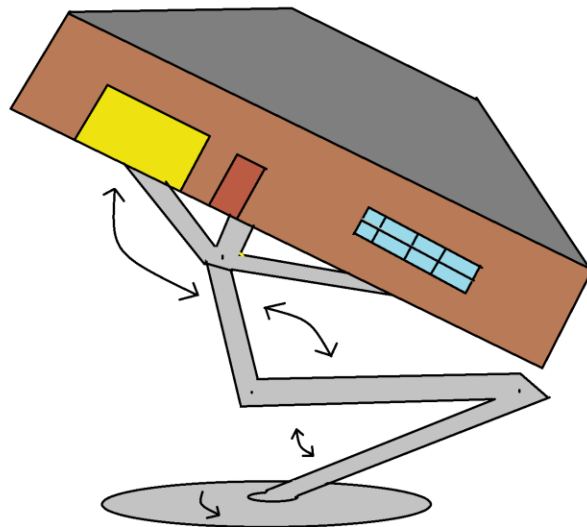
Characteristics of a vacuum tunnel that is capable of dropping objects at targeted (free fall) velocities

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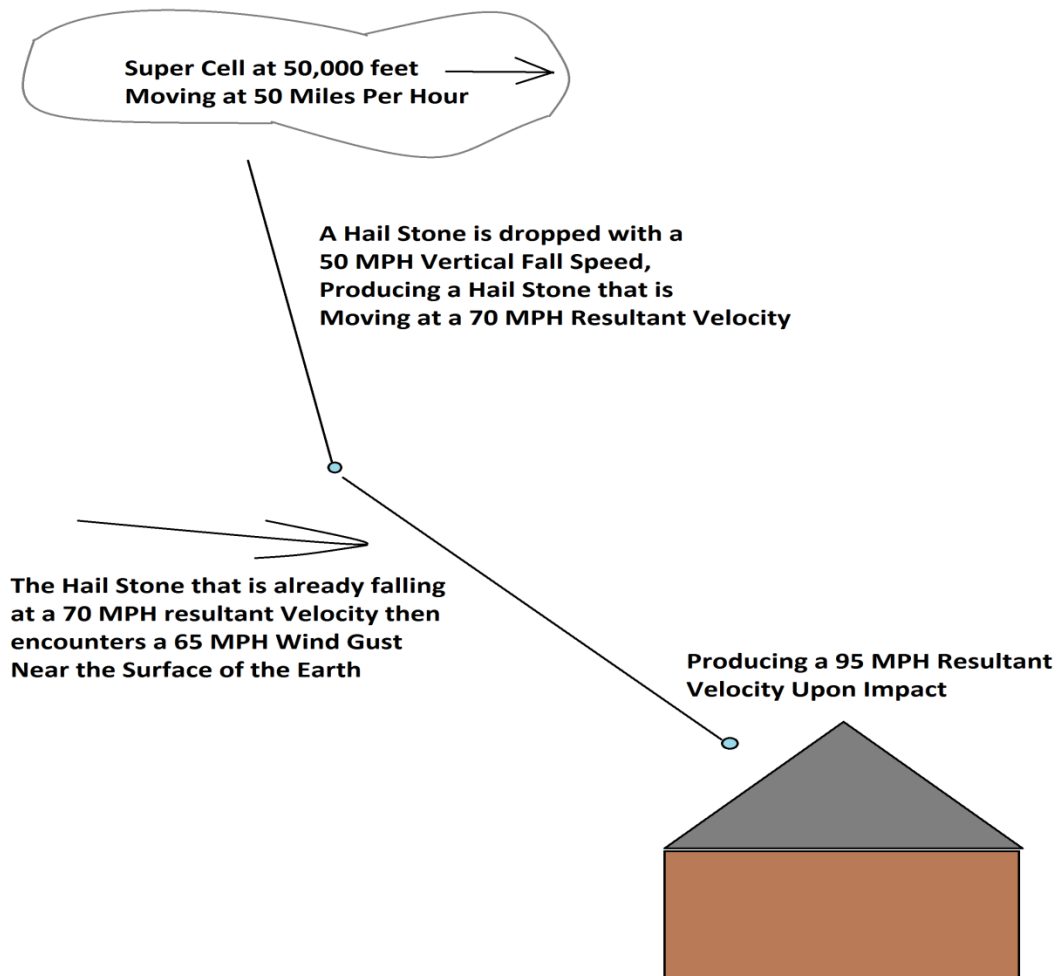


Now, as long as the hail stones were shaped into perfect spheres, sorted by size, and correctly placed into the specific vacuum tunnel of the correct length that would be necessary to enable the ice spheres to "Free Fall" to exactly the right targeted speed immediately prior to impact, then the only thing left to for us to worry about, would be to make sure that the re-sizing and sorting callander, and the 9 vacuum tunnels are all placed some how over a house that was built on some type of really fast rotating hydraulic machine that would enable the house to be twisted and rotated to the exact angle that would be required to enable the hail stones to impact each of the roof slopes at the perpendicular impact angles that would be required to replicate the laboraty testing scenarios at the Targeted "Free Fall Speeds" for each and every hail stone that would fall according to their pre-selected and specific sizes.



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How could a hail stone impact a sloped roof at a perpendicular angle of impact? Hail stones are produced by violent wind drafts that propel moisture molecules into the upper levels of the atmosphere where they become frozen. Hail stones cannot be formed without strong levels of wind that occur during severe thunderstorms. When the hail stones fall to the earth, they fall at “resultant” velocities that are caused by the motion of the thunderstorm as it passes over the earth, and from acceleration that occurs from wind gusts near the surface of the earth.



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Would wind driven hailstones impact roofing surfaces harder than free falling hailstones? In order to figure this out we'll need to know the mass of the hailstones first, and in order to figure out the mass of the hailstones we'll need to know what the volume is for 1", 1.25", 1.5", 1.75", 2", 2.25", 2.5", 2.75" & 3" diameter spheres.

According to page A-10 in the eighth edition of volume 1 of "Physics for Scientists and Engineers" written by Serway/Jewett, it states that the formula for determining the volume of a sphere is:
 $\frac{4}{3}\pi(\text{radius}^3)$

So the first half of the formula requires us to multiply $\frac{4}{3}$ x's π which gives us the number in red below.

$$\frac{4}{3} \times \pi = 4.1887902047863909846168578443727$$

Next in order to complete this formula we'll need to convert the radius of these spheres in to centimeters. To convert the radius of each sphere from inches to centimeters, the conversion chart on page A-1 of the eighth edition of volume 1 of "Physics for Scientists and Engineers" written by Serway/Jewett says that 1 centimeter equals .393733333 inches.

Using this conversion, the radius of each of the ice spheres can be calculated into the necessary the units that we'll need to complete the rest of this formula. (The radius of each sphere is equal to half the diameter of each sphere)

The second half of the equation is then shown below by multiplying the radius in centimeters by the power of 3.

The radius for a the radius for a 1" sphere is 1.27 cm $(1.27^3) = 2.048383$

the radius for a 1.25" sphere is 1.5875cm, $(1.5875^3) = 4.000748046875$

the radius for a 1.5" sphere is 1.905 cm, $(1.905^3) = 6.913292625$

the radius for a 1.75" sphere is 2.2225 cm, $(2.2225^3) = 10.978052640625$

the radius for a 2" sphere is 2.54 cm, $(2.54^3) = 16.387064$

the radius for a 2.25" sphere is 2.875cm, $(2.875^3) = 23.763671875$

the radius for a 2.5" sphere is 3.175cm, $(3.175^3) = 32.005984375$

the radius for a 2.75" sphere is 3.4925cm $(3.4925^3) = 42.599965203125$

and the radius for a 3" sphere is 3.81cm. $(3.81^3) = 55.306341$

Now, all that is left to be done to determine the volume of each sphere is to multiply the **red number** above $(\frac{4}{3} \times \pi)$ by the **blue numbers above** which are the (radius^3) .

Once the red number above is multiplied by each corresponding blue number, the volume for each sphere is then given in cubic centimeters.

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4.1887902047863909846168578443727 x's 2.048383 =The volume of a 1" sphere is 8.58024664cm

4.1887902047863909846168578443727 x's 4.000748046875 =The volume of a 1.25" sphere is 16.7582443cm

4.1887902047863909846168578443727 x's 6.913292625 =The volume of a 1.5" sphere ice 28.9583324 cm

4.1887902047863909846168578443727 x's 10.978052640625 =The volume of a 1.75" sphere is 45.984759 cm

4.1887902047863909846168578443727 x's 16.387064 =The volume of a 2" sphere is 68.64197316 cm

4.1887902047863909846168578443727 x's 23.763671875 =The volume of a 2.25" sphere is 97.73437195cm

4.1887902047863909846168578443727 x's 32.005984375 =The volume of a 2.5" sphere is 134.0663538cm

4.1887902047863909846168578443727 x's 42.599965203125 =The volume of a 2.75" sphere is 178.44231696cm

4.1887902047863909846168578443727 x's 55.306341 =The volume of a 3" sphere is 231.66665944 cm

Now that we've calculated the volume of the spheres, we're now ready to calculate the mass of the spheres. In order to calculate the mass of the sphere's we'll need to find out what the median density of ice is. The median density of ice can be found on page 404 of the eighth edition of volume 1 of "Physics for Scientists and Engineers" written by Serway/Jewett. The median density of ice is .917000

To find the mass of each individual ice sphere, the median density of Ice is multiplied by the volume of each sphere.

A 1" diameter ice sphere with a volume of 8.58024664cm multiplied by the median density of ice .917000 has a mass of 7.868 grams.

A 1.25" diameter ice sphere with a volume of 16.7582443cm multiplied by the median density of ice .917000 has a mass of 15.367355 grams.

A 1.5" diameter ice sphere with a volume of 28.9583324cm multiplied by the median density of ice .917000 has a mass of 26.5547 grams.

A 1.75" diameter ice sphere with a volume of 45.984759cm multiplied by the median density of ice .917000 has a mass of 42.168024 grams.

A 2" diameter ice sphere with a volume of 68.64197316cm multiplied by the median density of ice .917000 has a mass of 62.94468 grams.

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A 2.25" diameter ice sphere with a volume of 97.73437195cm multiplied by the median density of ice .917000 has a mass of 89.622419 grams.

A 2.5" diameter ice sphere with a volume of 134.0663538cm multiplied by the median density of ice .917000 has a mass of 122.9 grams.

A 2.75" diameter ice sphere with a volume of 178.44231696cm multiplied by the median density of ice .917000 has a mass of 163.631604 grams.

A 3" diameter ice sphere with a volume of 231.66665944cm multiplied by the median density of ice .917000 has a mass of 212.4383 grams.

In summary these calculations revealed the following masses for each of the corresponding ice spheres.

1" diameter ice spheres have a mass of 7.868 grams.
1.25" diameter ice spheres have a mass of 15.367355 grams.
1.5" diameter ice spheres have a mass of 26.5547 grams.
1.75" diameter ice spheres have a mass of 42.168024 grams.
2" diameter ice spheres have a mass of 62.94468 grams.
2.25" diameter ice spheres have a mass of 89.622419 grams.
2.5" diameter ice spheres have a mass of 122.9 grams.
2.75" diameter ice spheres have a mass of 163.631604 grams.
3" diameter ice spheres have a mass of 212.4383 grams.

Just to ensure that we're on the right track I then cross referenced our calculations with the calculations for the mass of 1 inch, 1.5 inch, 2 inch, 2.5 inch, & 3" diameter ice spheres that were determined by Vickie Crenshaw and Jim D. Koontz as published by them in their research publication entitled "Hail: Sizing it up!"

Vickie Crenshaw and Jim D. Koontz used a slightly lower value for the median density of ice than we did, which in their research publication was stated as being .91gm/cm³, and they published the following masses for the following 5 corresponding diameters of ice spheres.

Vickie Crenshaw and Jim D. Koontz found that:

1" diameter ice spheres have a mass of 7.85 grams, a difference of .018 grams
1.5" diameter ice spheres have a mass of 26.50 grams, a difference of .0547 grams
2" diameter ice spheres have a mass of 62.81 grams, a difference of .13468 grams
2.5" diameter ice spheres have a mass of 122.67 grams, a difference of .23 grams
3" diameter ice spheres have a mass of 211.98 grams, a difference of .4583 grams

Although Vickie Crenshaw and Jim D. Koontz did not offer any calculations for 1.25", 1.75", 2.25", or 2.75" inch diameter ice spheres, all of the other 5 masses correlated very well with our calculations and showed that only a negligible difference existed between their calculations and ours which was likely attributed to their use of a "rounded" median density of ice number that was used by them for their calculations which was .00700 gm/cm³ less than the .917000 median density of ice number we used from page 404 of the eighth edition of volume 1 of "Physics for Scientists and Engineers" written by Serway/Jewett.

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Now that we've calculated the masses for each of the ice spheres, the next order of business will be to determine how to calculate *resultant velocities*!

In order for us to calculate resultant velocities, we'll need to understand how to convert velocities from miles per hour, to meters per second. On the very last page of the eighth edition of Volume 1 "Physics for Scientists and Engineers" written by Serway/Jewett, it lists the conversion of 1 mile per hour being equal to 0.447 meters per second ^{^2}. Conversely one meter per second equals 2.23693629 miles per hour.

It will also be necessary to understand how to calculate the resultant velocity for the motion of a falling object that is being influenced by 2 vectors. On page 58 of the eighth edition of Volume 1 of "Physics for Scientists and Engineers" written by Serway/Jewett it shows that the Resultant vector is equal to the sum of vector A and vector B.

As an example, let's conceptualize how to go about calculating the resultant velocity of a falling object that is being influenced by a 56 mile per hour downward vertical force vector derived from gravity while simultaneously experiencing a 40 mile per hour horizontal wind gust vector as well.

First we'll need to convert 56 miles per hour to meters per second per second. So if we multiply 56 x 0.447 we find that 56 miles per hour is equal to 25.032 meters per second squared, or written differently 25.032^{^2}.

Next the 40 mile per hour wind gust will also need to be converted to meters per second as well. This is achieved by multiplying 40 x 0.447 which equals 17.88 meters per second squared, or written differently 40 miles per hour equals 17.88^{^2}.

Now to find the resultant velocity of these 2 vectors, we'll need to add (25.032^{^2}) + (17.88^{^2}), and then we'll need to take the square root of the sum.

So when we add (25.032^{^2}) + (17.88^{^2}) we find that the sum equals 946.295424, and now all that is left to do will be to take the square root of 946.295424, which equals 30.76191515 meters per second squared!

How fast is 30.76191515 meters per second squared? Well to find out how fast 30.76191515^{^2} is in miles per hour we'll need to multiply 30.76191515 x's 2.23693629 which equals 68.812, or 68.812 miles per hour. Therefore the resultant velocity from both the influence of gravity @56 mph vertical, and the influence of a 40 mile per hour horizontal wind gust would produce a resultant velocity of 68.813 miles per hour.

So now that we've figured out how to calculate the volume of the ice spheres, and the mass of the ice spheres, and the resultant velocity of vertical and horizontal vectors of force, the only thing left to determine is how much kinetic energy these hailstones will produce during their periods of accelerated motion.

To figure this out, the formula for Kinetic energy will be needed. On Page 100 of "Elementary Physics, Atoms, Waves, Particles" written by George A Williams, this formula for determining Kinetic energy is provided as shown below.

Kinetic energy equals: $\frac{1}{2} \times \text{x's mass} \times \text{x's (velocity)}^2$

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So for example, if we desired to calculate the kinetic energy of a 2 inch diameter ice sphere traveling at a 101.242 mile per hour resultant velocity, we'd need to know the answer to these 2 questions before we can begin crunching in the numbers to finding the ice spheres kinetic energy.

Question1: **What is the mass of a 2 inch diameter ice sphere?** Answer1: **62.94468 grams.**

Question2: **What is 101.242 miles per hour equal to in meters per second?** Answer2: **45.255174 m/s/s**

Now we simply plug these numbers into the kinetic energy formula: $\frac{1}{2} \times \text{mass} \times (\text{velocity})^2$ as shown below:

$$(.5)(.06294468)(45.255174^2)=64.456 \text{ Joules}$$

Thus the answer is that a 2 inch diameter sphere of ice traveling at a 101.242 mile per hour resultant velocity will produce 64.45 joules of kinetic energy.

Below please find some additional calculations that were made to find the energy levels that could be achieved with real hail stones of like masses being propelled at resultant velocities that would be reasonably attainable during real life hail storm scenarios. The results are referenced below.

A 1 inch diameter sphere of ice with a 1.27 cm radius, a volume of 8.58024664cm³, and a mass of 7.868 grams would achieve the following energy levels depending on its resultant velocities.

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
(.5*.007868)*(25.2884²)= 2.51581 Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
(.5*.007868)*(26.91536²)= 2.84993 Joules
45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
(.5*.007868)*(28.4494²)= 3.18406 Joules
40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
(.5*.007868)*(28.62453²)= 3.22338 Joules
45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
(.5*.007868)*(30.07153²)= 3.5575 Joules
40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
(.5*.007868)*(30.40201²)= 3.63613 Joules
50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
(.5*.007868)*(31.6105²)= 3.93095 Joules
45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
(.5*.007868)*(31.76815²)= 3.97025 Joules
40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
(.5*.007868)*(32.2361²)= 4.08808 Joules
50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
(.5*.007868)*(33.22864²)= 4.3437 Joules
45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
(.5*.007868)*(33.52799²)= 4.42231 Joules
40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
(.5*.007868)*(34.1185²)= 4.57946 Joules
55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
(.5*.007868)*(34.77155²)= 4.75644 Joules
50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
(.5*.007868)*(34.91494²)= 4.79575 Joules
45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
(.5*.007868)*(35.34161²)= 4.91368 Joules
40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
(.5*.007868)*(36.04152²)= 5.11023 Joules
55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
(.5*.007868)*(36.30642²)= 5.18563 Joules

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50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
(.5*.007868)*(36.66001^2) = 5.28712 Joules

45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
(.5*.007868)*(37.20114^2) = 5.44436 Joules

60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
(.5*.007868)*(37.9362^2) = 5.66164 Joules

40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
(.5*.007868)*(37.9984^2) = 5.68022 Joules

55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
(.5*.007868)*(38.0640^2) = 5.69985 Joules

50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
(.5*.007868)*(38.45583^2) = 5.8178 Joules

45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
(.5*.007868)*(39.10003^2) = 6.01435 Joules

60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
(.5*.007868)*(39.54472^2) = 6.15193 Joules

55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
(.5*.007868)*(39.7966^2) = 6.23055 Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
(.5*.007868)*(39.98447^2) = 6.28951 Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
(.5*.007868)*(40.29564^2) = 6.38779 Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
(.5*.007868)*(41.03281^2) = 6.62364 Joules

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
(.5*.007868)*(41.09365^2) = 6.6433 Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
(.5*.007868)*(41.21505^2) = 6.68261 Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
(.5*.007868)*(41.57712^2) = 6.80054 Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
(.5*.007868)*(41.9955^2) = 6.93809 Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
(.5*.007868)*(42.17367^2) = 6.99708 Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
(.5*.007868)*(42.7034^2) = 7.17397 Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
(.5*.007868)*(42.93679^2) = 7.2526 Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
(.5*.007868)*(42.994^2) = 7.27194 Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
(.5*.007868)*(44.08503^2) = 7.64569 Joules

55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
(.5*.007868)*(44.2274^2) = 7.69515 Joules

70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
(.5*.007868)*(44.2547^2) = 7.70465 Joules

65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
(.5*.007868)*(44.36745^2) = 7.74396 Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
(.5*.007868)*(44.704^2) = 7.86189 Joules

55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
(.5*.007868)*(45.25935^2) = 8.05844 Joules

70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
(.5*.007868)*(45.86247^2) = 8.27464 Joules

65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
(.5*.007868)*(46.07983^2) = 8.35326 Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
(.5*.007868)*(46.5115^2) = 8.5105 Joules

75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
(.5*.007868)*(47.41575^2) = 8.84463 Joules

70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
(.5*.007868)*(47.521^2) = 8.88394 Joules

65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
(.5*.007868)*(47.83537^2) = 9.00187 Joules

75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
(.5*.007868)*(49.02176^2) = 9.45393 Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
(.5*.007868)*(49.22517^2) = 9.53254 Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
(.5*.007868)*(50.5768^2) = 9.53254 Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
(.5*.007868)*(50.67549^2) = 10.10253 Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
(.5*.007868)*(52.18127^2) = 10.71183 Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
(.5*.007868)*(53.73785^2) = 11.36043 Joules

A 1.25" diameter ice sphere with a 1.5875cm radius, a volume of 16.7582443cm³, and a density of 15.367355 grams would achieve the following energy levels depending on its resultant velocities

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56
(.5*.0153673)*(25.2884^2) = 4.9137 Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
(.5*.0153673)*(26.91536^2) = 5.565562901012 Joules
45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
(.5*.0153673)*(28.4494^2) = 6.2189 Joules
40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
(.5*.0153673)*(28.62453^2) = 6.2957 Joules
45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
(.5*.0153673)*(30.07153^2) = 6.94830 Joules
40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
(.5*.0153673)*(30.40201^2) = 7.1018 Joules
50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
(.5*.0153673)*(31.6105^2) = 7.6776852612624 Joules
45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
(.5*.0153673)*(31.76815^2) = 7.754457558008 Joules
40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
(.5*.0153673)*(32.2361^2) = 7.9845889362755 Joules
50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
(.5*.0153673)*(33.22864^2) = 8.4838446449 Joules
45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
(.5*.0153673)*(33.52799^2) = 8.637391611534 Joules
40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
(.5*.0153673)*(34.1185^2) = 8.944322147434 Joules
55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
(.5*.0153673)*(34.77155^2) = 9.28999916612 Joules
50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
(.5*.0153673)*(34.91494^2) = 9.3667 Joules
45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
(.5*.0153673)*(35.34161^2) = 9.5971047292 Joules
40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
(.5*.0153673)*(36.04152^2) = 9.9809 Joules
55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
(.5*.0153673)*(36.30642^2) = 10.12825037298 Joules
50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
(.5*.0153673)*(36.66001^2) = 10.32649007959 Joules
45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
(.5*.0153673)*(37.20114^2) = 10.6335939224 Joules
60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
(.5*.0153673)*(37.9362^2) = 11.057965393716 Joules
40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
(.5*.0153673)*(37.9984^2) = 11.09425628783 Joules
55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
(.5*.0153673)*(38.064^2) = 11.13259534583 Joules
50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
(.5*.0153673)*(38.45583^2) = 11.3629724180 Joules
45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
(.5*.0153673)*(39.10003^2) = 11.7468589823 Joules
60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
(.5*.0153673)*(39.54472^2) = 12.01557569227 Joules
55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
(.5*.0153673)*(39.7966^2) = 12.169129531787 Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.0153673 \times (39.98447)^2 = 12.284295686$ Joules
50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.0153673 \times (40.29564)^2 = 12.476239117$ Joules
45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.0153673 \times (41.03281)^2 = 12.93689616705$ Joules
65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 \times 0.0153673 \times (41.09365)^2 = 12.975288091533$ Joules
60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.0153673 \times (41.21505)^2 = 13.052065244404$ Joules
55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.0153673 \times (41.57712)^2 = 13.282394647269$ Joules
40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.0153673 \times (41.9955)^2 = 13.551054335894$ Joules
50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.0153673 \times (42.17367)^2 = 13.66628158625$ Joules
65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.0153673 \times (42.7034)^2 = 14.011753321937$ Joules
60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.0153673 \times (42.93679)^2 = 14.165330767636$ Joules
45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.0153673 \times (42.994)^2 = 14.203104363211$ Joules
50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.0153673 \times (44.08503)^2 = 14.933095940401$ Joules
55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 \times 0.0153673 \times (44.2274)^2 = 15.029702784261$ Joules
70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.0153673 \times (44.2547)^2 = 15.048263112074$ Joules
65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.0153673 \times (44.36745)^2 = 15.12503927554$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.0153673 \times (44.704)^2 = 15.355372024678$ Joules
55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 \times 0.0153673 \times (45.25935)^2 = 15.73925598738$ Joules
70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.0153673 \times (45.86247)^2 = 16.161529353031$ Joules
65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.0153673 \times (46.07983)^2 = 16.3150838583$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.0153673 \times (46.5115)^2 = 16.622190892338$ Joules
75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.0153673 \times (47.41575)^2 = 17.27479183784$ Joules
70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.0153673 \times (47.521)^2 = 17.35156758274$ Joules
65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.0153673 \times (47.83537)^2 = 17.5819017574$ Joules
75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.0153673 \times (49.02176)^2 = 18.46483251814$ Joules
70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.0153673 \times (49.22517)^2 = 18.61838571491$ Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.0153673 \times (50.5768)^2 = 19.654874268832$ Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.0153673 \times (50.67549)^2 = 19.731653821$ Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.0153673 \times (52.18127)^2 = 20.9216948601$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.0153673 \times (53.73785)^2 = 22.18851040504$ Joules

A 1.5 inch diameter sphere of ice with a 1.905 cm radius, and a volume of 28.9583324 cm³, and a density of 26.5547 grams would achieve the following energy levels dependent upon its resultant velocities.

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 \times 0.0265547 \times (25.2884)^2 = 8.4909074747442$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
 $0.5 \times 0.0265547 \times (26.91536)^2 = 9.6185983431847$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
 $0.5 \times 0.0265547 \times (28.4494)^2 = 10.746266999426$ Joules

40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
 $0.5 \times 0.0265547 \times (28.62453)^2 = 10.87897885748$ Joules

45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
 $0.5 \times 0.0265547 \times (30.07153)^2 = 12.006666664834$ Joules

40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
 $0.5 \times 0.0265547 \times (30.40201)^2 = 12.272018428031$ Joules

50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
 $0.5 \times 0.0265547 \times (31.6105)^2 = 13.267042929288$ Joules

45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
 $0.5 \times 0.0265547 \times (31.76815)^2 = 13.399705486042$ Joules

40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
 $0.5 \times 0.0265547 \times (32.2361)^2 = 13.797372591549$ Joules

50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
 $0.5 \times 0.0265547 \times (33.22864)^2 = 14.660086638127$ Joules

45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
 $0.5 \times 0.0265547 \times (33.52799)^2 = 14.925415852284$ Joules

40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
 $0.5 \times 0.0265547 \times (34.1185)^2 = 15.455791930168$ Joules

55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
 $0.5 \times 0.0265547 \times (34.77155)^2 = 16.053121944438$ Joules

50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
 $0.5 \times 0.0265547 \times (34.91494)^2 = 16.185793816961$ Joules

45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
 $0.5 \times 0.0265547 \times (35.34161)^2 = 16.583800469464$ Joules

40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
 $0.5 \times 0.0265547 \times (36.04152)^2 = 17.247160330146$ Joules

55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
 $0.5 \times 0.0265547 \times (36.30642)^2 = 17.247160330146$ Joules

50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
 $0.5 \times 0.0265547 \times (36.66001)^2 = 17.844178620614$ Joules

45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
 $0.5 \times 0.0265547 \times (37.20114)^2 = 18.374854172973$ Joules

60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
 $0.5 \times 0.0265547 \times (37.9362)^2 = 19.108168229977$ Joules

40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
 $0.5 \times 0.0265547 \times (37.9984)^2 = 19.17087890823$ Joules

55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
 $0.5 \times 0.0265547 \times (38.064)^2 = 19.237128814426$ Joules

50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
 $0.5 \times 0.0265547 \times (38.45583)^2 = 19.635220479151$ Joules

45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
 $0.5 \times 0.0265547 \times (39.10003)^2 = 20.29857660217$ Joules

60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
 $0.5 \times 0.0265547 \times (39.54472)^2 = 20.762919174853$ Joules

55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
 $0.5 \times 0.0265547 \times (39.7966)^2 = 21.028260265482$ Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.0265547 \times (39.98447)^2 = 21.22726742260$ Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.0265547 \times (40.29564)^2 = 21.55894574067$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.0265547 \times (41.03281)^2 = 22.35496129100$ Joules

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 \times 0.0265547 \times (41.09365)^2 = 22.421302550496$ Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.0265547 \times (41.21505)^2 = 22.553973498635$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.0265547 \times (41.57712)^2 = 22.951982790721$ Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.0265547 \times (41.9955)^2 = 23.416226830566$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.0265547 \times (42.17367)^2 = 23.61533956118$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.0265547 \times (42.7034)^2 = 24.212314846332$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.0265547 \times (42.93679)^2 = 24.477696728465$ Joules
45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.0265547 \times (42.994)^2 = 24.542969515385$ Joules
50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.0265547 \times (44.08503)^2 = 25.804395226784$ Joules
55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 \times 0.0265547 \times (44.08503)^2 = 25.804395226784$ Joules
70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.0265547 \times (44.2547)^2 = 26.003404141404$ Joules
65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.0265547 \times (44.36745)^2 = 26.136073379852$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.0265547 \times (44.704)^2 = 26.534088454298$ Joules
55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 \times 0.0265547 \times (45.25935)^2 = 27.19744008175$ Joules
70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.0265547 \times (45.86247)^2 = 27.927128611463$ Joules
65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.0265547 \times (46.07983)^2 = 28.1924708525$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.0265547 \times (46.5115)^2 = 28.723151919255$ Joules
75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.0265547 \times (46.5115)^2 = 28.723151919255$ Joules
70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.0265547 \times (47.521)^2 = 29.983515106061$ Joules
65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.0265547 \times (47.83537)^2 = 30.3815326439$ Joules
75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.0265547 \times (49.02176)^2 = 31.90723732012$ Joules
70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.0265547 \times (49.22517)^2 = 32.17257730009$ Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.0265547 \times (50.5768)^2 = 33.963629898977$ Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.0265547 \times (50.67549)^2 = 34.0963049938$ Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.0265547 \times (52.18127)^2 = 36.15269634234$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.0265547 \times (53.73785)^2 = 38.34175406564$ Joules

A 1.75 inch diameter sphere of ice with a radius of 2.2225 cm, and a volume of 45.984759 cm³, and a density of 42.16802433 grams would achieve the following energy levels dependent upon its resultant velocities.

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 \times 0.04216802 \times (25.2884)^2 = 13.483291327455$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
 $0.5 \times 0.04216802 \times (26.91536)^2 = 15.274028601618$ Joules
45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
 $0.5 \times 0.04216802 \times (28.4494)^2 = 17.064730603514$ Joules
40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
 $0.5 \times 0.04216802 \times (28.62453)^2 = 17.2754728180$ Joules
45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
 $0.5 \times 0.04216802 \times (30.07153)^2 = 19.06620523131$ Joules
40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
 $0.5 \times 0.04216802 \times (30.40201)^2 = 19.487575401476$ Joules
50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
 $0.5 \times 0.04216802 \times (31.6105)^2 = 21.067642699148$ Joules
45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
 $0.5 \times 0.04216802 \times (31.76815)^2 = 21.278306624798$ Joules
40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
 $0.5 \times 0.04216802 \times (32.2361)^2 = 21.909789355101$ Joules
50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
 $0.5 \times 0.04216802 \times (33.22864)^2 = 23.27975185403$ Joules
45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
 $0.5 \times 0.04216802 \times (33.52799)^2 = 23.701086217032$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
 $0.5 \times 0.04216802 \times (34.1185)^2 = 24.543306579519$ Joules

55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
 $0.5 \times 0.04216802 \times (34.77155)^2 = 25.49184766596$ Joules

50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
 $0.5 \times 0.04216802 \times (34.91494)^2 = 25.70252638476$ Joules

45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
 $0.5 \times 0.04216802 \times (35.34161)^2 = 26.3345483049$ Joules

40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
 $0.5 \times 0.04216802 \times (36.04152)^2 = 27.38794268979$ Joules

55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
 $0.5 \times 0.04216802 \times (36.30642)^2 = 27.792017094296$ Joules

50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
 $0.5 \times 0.04216802 \times (36.66001)^2 = 28.335988768754$ Joules

45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
 $0.5 \times 0.04216802 \times (37.20114)^2 = 29.17868468719$ Joules

60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
 $0.5 \times 0.04216802 \times (37.9362)^2 = 30.34316411351$ Joules

40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
 $0.5 \times 0.04216802 \times (37.9984)^2 = 30.442746678359$ Joules

55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
 $0.5 \times 0.04216802 \times (38.064)^2 = 30.547949424745$ Joules

50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
 $0.5 \times 0.04216802 \times (38.45583)^2 = 31.18010634159$ Joules

45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
 $0.5 \times 0.04216802 \times (39.10003)^2 = 32.23349479120$ Joules

60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
 $0.5 \times 0.04216802 \times (39.54472)^2 = 32.970856045205$ Joules

55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
 $0.5 \times 0.04216802 \times (39.7966)^2 = 33.392209267665$ Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.04216802 \times (39.98447)^2 = 33.7082263110$ Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.04216802 \times (40.29564)^2 = 34.23492094324$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.04216802 \times (41.03281)^2 = 35.4989683490$ Joules

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 \times 0.04216802 \times (41.09365)^2 = 35.60431616156$ Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.04216802 \times (41.21505)^2 = 35.814993412462$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.04216802 \times (41.57712)^2 = 36.447019524181$ Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.04216802 \times (41.9955)^2 = 37.184224311171$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.04216802 \times (42.17367)^2 = 37.5004090018$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.04216802 \times (42.7034)^2 = 38.448386789775$ Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.04216802 \times (42.93679)^2 = 38.869804787848$ Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.04216802 \times (42.994)^2 = 38.973455899864$ Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.04216802 \times (44.08503)^2 = 40.97655985610$ Joules

55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 \times 0.04216802 \times (44.2274)^2 = 41.241649971093$ Joules

70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.04216802 \times (44.2547)^2 = 41.29257969033$ Joules

65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.04216802 \times (44.36745)^2 = 41.503254226297$ Joules

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 $0.5 \times 0.04216802 \times (44.704)^2 = 42.13528952022$ Joules

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 $0.5 \times 0.04216802 \times (45.25935)^2 = 43.1886708310$ Joules

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A 2 inch diameter ice sphere with a 2.54 cm radius, and a volume of 68.64197316 cm³, and a density of 62.94468 grams would achieve the following resultant velocities

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 $0.5 \times 0.06294468 \times (39.7966)^2 = 49.844928143323$ Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.06294468 \times (39.98447)^2 = 50.316650355311$ Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.06294468 \times (40.29564)^2 = 51.102853385043$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.06294468 \times (41.03281)^2 = 52.989711232834$ Joules

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 \times 0.06294468 \times (41.09365)^2 = 52.989711232834$ Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.06294468 \times (41.21505)^2 = 53.461445416444$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.06294468 \times (41.57712)^2 = 54.404877936012$ Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.06294468 \times (41.9955)^2 = 55.505311852795$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.06294468 \times (42.17367)^2 = 55.977284313885$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.06294468 \times (42.7034)^2 = 57.392341471063$ Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.06294468 \times (42.7034)^2 = 57.392341471063$ Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.06294468 \times (42.994)^2 = 58.176118065564$ Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.06294468 \times (44.08503)^2 = 61.166173978371$ Joules

55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 \times 0.06294468 \times (44.2274)^2 = 61.561876988828$ Joules

70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.06294468 \times (44.2547)^2 = 61.637900356297$ Joules

65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.06294468 \times (44.36745)^2 = 61.952376616993$ Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.06294468 \times (44.704)^2 = 62.895822842941$ Joules

55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 \times 0.06294468 \times (45.25935)^2 = 64.46821702994$ Joules

70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.06294468 \times (45.86247)^2 = 66.197854758945$ Joules

65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.06294468 \times (46.07983)^2 = 66.82681620284$ Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.06294468 \times (46.5115)^2 = 68.084730994847$ Joules

75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.06294468 \times (47.41575)^2 = 70.757793776361$ Joules

70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.06294468 \times (47.521)^2 = 71.072268322602$ Joules

65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.06294468 \times (47.83537)^2 = 72.015720387909$ Joules

75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.06294468 \times (49.02176)^2 = 75.632217377681$ Joules

70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.06294468 \times (49.22517)^2 = 76.26117346194$ Joules

80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.06294468 \times (50.5768)^2 = 80.506645363327$ Joules

75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.06294468 \times (50.67549)^2 = 80.821135506082$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.06294468 \times (52.18127)^2 = 85.695560575199$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.06294468 \times (53.73785)^2 = 90.884455117193$ Joules

A 2.25 inch diameter sphere of ice with a 2.875cm radius, and a volume of 97.73437 195 cm³, and a density of 89.622419 grams would achieve the following energy levels at the following resultant velocities

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 \times 0.08962241 \times (25.2884)^2 = 28.656907853359$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
 $0.5 \times 0.08962241 \times (26.91536)^2 = 32.462877168193$ Joules
45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
 $0.5 \times 0.08962241 \times (28.4494)^2 = 36.268771516606$ Joules
40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
 $0.5 \times 0.08962241 \times (28.62453)^2 = 36.716675524353$ Joules
45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
 $0.5 \times 0.08962241 \times (30.07153)^2 = 40.522634507982$ Joules
40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
 $0.5 \times 0.08962241 \times (30.40201)^2 = 41.418199681582$ Joules
50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
 $0.5 \times 0.08962241 \times (31.6105)^2 = 44.776418520873$ Joules
45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
 $0.5 \times 0.08962241 \times (31.76815)^2 = 45.224156136174$ Joules
40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
 $0.5 \times 0.08962241 \times (32.2361)^2 = 46.566287072443$ Joules
50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
 $0.5 \times 0.08962241 \times (33.22864)^2 = 49.477956644877$ Joules
45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
 $0.5 \times 0.08962241 \times (33.52799)^2 = 50.373445715218$ Joules
40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
 $0.5 \times 0.08962241 \times (34.1185)^2 = 52.163470920033$ Joules
55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
 $0.5 \times 0.08962241 \times (34.77155)^2 = 54.179466410257$ Joules
50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
 $0.5 \times 0.08962241 \times (34.91494)^2 = 54.627235466381$ Joules
45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
 $0.5 \times 0.08962241 \times (35.34161)^2 = 55.970512377564$ Joules
40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
 $0.5 \times 0.08962241 \times (36.04152)^2 = 58.209359339178$ Joules
55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
 $0.5 \times 0.08962241 \times (36.30642)^2 = 59.068164707567$ Joules
50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
 $0.5 \times 0.08962241 \times (36.66001)^2 = 60.224302758078$ Joules
45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
 $0.5 \times 0.08962241 \times (37.20114)^2 = 62.0153386926$ Joules
60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
 $0.5 \times 0.08962241 \times (37.9362)^2 = 64.490281850517$ Joules
40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
 $0.5 \times 0.08962241 \times (37.9984)^2 = 64.701931092189$ Joules
55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
 $0.5 \times 0.08962241 \times (38.064)^2 = 64.925525267816$ Joules
50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
 $0.5 \times 0.08962241 \times (38.45583)^2 = 66.2690890962$ Joules
45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
 $0.5 \times 0.08962241 \times (39.10003)^2 = 68.507923443177$ Joules
60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
 $0.5 \times 0.08962241 \times (39.54472)^2 = 70.075084828131$ Joules
55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
 $0.5 \times 0.08962241 \times (39.7966)^2 = 70.970613981696$ Joules
40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.08962241 \times (39.98447)^2 = 71.642265366515$ Joules
50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.08962241 \times (40.29564)^2 = 72.761683405877$ Joules
45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.08962241 \times (41.03281)^2 = 75.448244806244$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 * 0.08962241 * (41.09365)^2 = 75.672147300276$ Joules
60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 * 0.08962241 * (41.21505)^2 = 76.119913236595$ Joules
55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 * 0.08962241 * (41.57712)^2 = 77.463199056398$ Joules
40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 * 0.08962241 * (41.9955)^2 = 79.030027891937$ Joules
50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 * 0.08962241 * (42.17367)^2 = 79.702035588481$ Joules
65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 * 0.08962241 * (42.7034)^2 = 81.716833863951$ Joules
60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 * 0.08962241 * (42.93679)^2 = 82.612500689301$ Joules
45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 * 0.08962241 * (42.994)^2 = 82.832797076423$ Joules
50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 * 0.08962241 * (44.08503)^2 = 87.090122984515$ Joules
55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 * 0.08962241 * (44.2274)^2 = 87.653536086963$ Joules
70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 * 0.08962241 * (44.2547)^2 = 87.761780300912$ Joules
65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 * 0.08962241 * (44.36745)^2 = 88.209540467004$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 * 0.08962241 * (44.704)^2 = 89.552845802337$ Joules
55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 * 0.08962241 * (45.25935)^2 = 91.791664976711$ Joules
70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 * 0.08962241 * (45.86247)^2 = 94.254371939402$ Joules
65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 * 0.08962241 * (46.07983)^2 = 95.149904975696$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 * 0.08962241 * (46.5115)^2 = 96.940959521279$ Joules
75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 * 0.08962241 * (47.41575)^2 = 100.74694167197$ Joules
70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 * 0.08962241 * (47.521)^2 = 101.19469939697$ Joules
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 $0.5 * 0.08962241 * (47.83537)^2 = 102.53801304654$ Joules
75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 * 0.08962241 * (49.02176)^2 = 107.68728342144$ Joules
70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 * 0.08962241 * (49.22517)^2 = 108.58280882653$ Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 * 0.08962241 * (50.5768)^2 = 114.62763141344$ Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 * 0.08962241 * (50.67549)^2 = 115.07541134519$ Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 * 0.08962241 * (52.18127)^2 = 122.01575518456$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 * 0.08962241 * (53.73785)^2 = 129.40384952532$ Joules

A 2.5 inch diameter sphere of ice with a 3.175cm radius, and a volume of 134.0663538cm³, and a density of 122.92884 grams would achieve the following energy levels at the following resultant velocities

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 * 0.12293884 * (25.2884)^2 = 39.309889228362$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
 $0.5 * 0.12293884 * (26.91536)^2 = 44.530697870322$ Joules
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 $0.5 \times 0.12293884 \times (30.40201)^2 = 56.815091490422$ Joules

50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
 $0.5 \times 0.12293884 \times (31.6105)^2 = 61.421701919316$ Joules

45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
 $0.5 \times 0.12293884 \times (31.76815)^2 = 62.035882491446$ Joules

40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
 $0.5 \times 0.12293884 \times (32.2361)^2 = 63.876940106756$ Joules

50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
 $0.5 \times 0.12293884 \times (33.22864)^2 = 67.871000071203$ Joules

45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
 $0.5 \times 0.12293884 \times (33.52799)^2 = 69.099380200017$ Joules

40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
 $0.5 \times 0.12293884 \times (34.1185)^2 = 71.554833275323$ Joules

55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
 $0.5 \times 0.12293884 \times (34.77155)^2 = 74.320259322372$ Joules

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 $0.5 \times 0.12293884 \times (34.91494)^2 = 74.934483023205$ Joules

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 $0.5 \times 0.12293884 \times (35.34161)^2 = 76.777112620642$ Joules

40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
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50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
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 $0.5 \times 0.12293884 \times (39.10003)^2 = 93.975208197515$ Joules

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 $0.5 \times 0.12293884 \times (39.7966)^2 = 97.353384683558$ Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.12293884 \times (39.98447)^2 = 98.274717217842$ Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.12293884 \times (40.29564)^2 = 99.81027015861$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.12293884 \times (41.03281)^2 = 103.49553974855$ Joules

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 $0.5 \times 0.12293884 \times (41.21505)^2 = 104.41689566491$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.12293884 \times (41.57712)^2 = 106.25953748267$ Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.12293884 \times (41.9955)^2 = 108.408822684$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.12293884 \times (42.17367)^2 = 109.3306439861$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.12293884 \times (42.7034)^2 = 112.09442776318$ Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.12293884 \times (42.93679)^2 = 113.32305172603$ Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.12293884 \times (42.994)^2 = 113.62524157218$ Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.12293884 \times (44.08503)^2 = 119.46519509098$ Joules

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70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.12293884 \times (44.2547)^2 = 120.38653576186$ Joules
65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.12293884 \times (44.36745)^2 = 121.00074726786$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.12293884 \times (44.704)^2 = 122.8434158559$ Joules
55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 \times 0.12293884 \times (45.25935)^2 = 125.91449854903$ Joules
70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.12293884 \times (45.86247)^2 = 129.2926975648$ Joules
65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.12293884 \times (46.07983)^2 = 130.52113800357$ Joules
60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.12293884 \times (46.5115)^2 = 132.97800306902$ Joules
75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.12293884 \times (47.41575)^2 = 138.19882931846$ Joules
70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.12293884 \times (47.521)^2 = 138.81303747591$ Joules
65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.12293884 \times (47.83537)^2 = 140.65571746896$ Joules
75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.12293884 \times (49.02176)^2 = 147.71918883438$ Joules
70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.12293884 \times (49.22517)^2 = 148.94761880511$ Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.12293884 \times (50.5768)^2 = 157.23955691345$ Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.12293884 \times (50.67549)^2 = 157.85379553285$ Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.12293884 \times (52.18127)^2 = 167.37415791556$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.12293884 \times (53.73785)^2 = 177.50871854682$ Joules

A 2.75 inch diameter sphere of ice with a 3.4925cm radius, and a volume of 178.44231696cm³, and a density of 163.631604 gram s would achieve the following energy levels at the following resultant velocities

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 \times 0.1636316 \times (25.2884)^2 = 52.321463829166$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
 $0.5 \times 0.1636316 \times (26.91536)^2 = 59.270360299783$ Joules
45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
 $0.5 \times 0.1636316 \times (28.4494)^2 = 66.219119897542$ Joules
40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
 $0.5 \times 0.1636316 \times (28.62453)^2 = 67.03689805631$ Joules
45 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 30.07153 meters per second or to 67.2680 mph.
 $0.5 \times 0.1636316 \times (30.07153)^2 = 73.985775664327$ Joules
40 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 30.40201 meters per second or to 68.027 mph.
 $0.5 \times 0.1636316 \times (30.40201)^2 = 75.62088860383$ Joules
50 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 31.6105 meters per second or to 70.7106 mph.
 $0.5 \times 0.1636316 \times (31.6105)^2 = 81.752287233072$ Joules
45 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 31.76815 meters per second or to 71.063 mph.
 $0.5 \times 0.1636316 \times (31.76815)^2 = 82.56976159436$ Joules
40 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 32.2361 meters per second or to 72.111 mph.
 $0.5 \times 0.1636316 \times (32.2361)^2 = 85.020209339641$ Joules
50 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 33.22864 meters per second or to 74.3303 mph.
 $0.5 \times 0.1636316 \times (33.22864)^2 = 90.336303280974$ Joules
45 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 33.52799 meters per second or to 74.999 mph.
 $0.5 \times 0.1636316 \times (33.52799)^2 = 91.971277271993$ Joules
40 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 34.1185 meters per second or to 76.32169 mph.
 $0.5 \times 0.1636316 \times (34.1185)^2 = 95.239485394318$ Joules
55 mph vertical velocity with a 55 mph cross wind gust would increase the velocity to 34.77155 meters per second or to 77.7817 mph.
 $0.5 \times 0.1636316 \times (34.77155)^2 = 98.920267552017$ Joules
50 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 34.91494 meters per second or to 78.1024 mph.
 $0.5 \times 0.1636316 \times (34.91494)^2 = 99.737799317611$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

45 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 35.34161 meters per second or to 79.05693 mph.
 $0.5 \times 0.1636316 \times (35.34161)^2 = 102.19033937115$ Joules

40 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 36.04152 meters per second or to 80.6226 mph.
 $0.5 \times 0.1636316 \times (36.04152)^2 = 106.27800126826$ Joules

55 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 36.30642 meters per second or to 81.394 mph.
 $0.5 \times 0.1636316 \times (36.30642)^2 = 107.84599856401$ Joules

50 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 36.66001 meters per second or to 82.066 mph.
 $0.5 \times 0.1636316 \times (36.66001)^2 = 109.95686256583$ Joules

45 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 37.20114 meters per second or to 83.2165 mph.
 $0.5 \times 0.1636316 \times (37.20114)^2 = 113.22691606722$ Joules

60 mph vertical velocity with a 60 mph cross wind gust would increase the velocity to 37.9362 meters per second or to 84.8528 mph.
 $0.5 \times 0.1636316 \times (37.9362)^2 = 117.74563977526$ Joules

40 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 37.9984 meters per second or to 84.999 mph.
 $0.5 \times 0.1636316 \times (37.9984)^2 = 118.13206660817$ Joules

55 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 38.0640 meters per second or to 85.0640 mph.
 $0.5 \times 0.1636316 \times (38.064)^2 = 118.54030236872$ Joules

50 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 38.45583 meters per second or to 86.0232 mph.
 $0.5 \times 0.1636316 \times (38.45583)^2 = 120.9933662725$ Joules

45 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 39.10003 meters per second or to 87.4642 mph.
 $0.5 \times 0.1636316 \times (39.10003)^2 = 125.08100513794$ Joules

60 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 39.54472 meters per second or to 88.459 mph.
 $0.5 \times 0.1636316 \times (39.54472)^2 = 127.94231097516$ Joules

55 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 39.7966 meters per second or to 89.022 mph.
 $0.5 \times 0.1636316 \times (39.7966)^2 = 129.57735814968$ Joules

40 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 39.98447 meters per second or to 89.4427 mph.
 $0.5 \times 0.1636316 \times (39.98447)^2 = 130.80365178249$ Joules

50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.1636316 \times (40.29564)^2 = 132.84747279611$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.1636316 \times (41.03281)^2 = 137.75256673903$ Joules

65 mph vertical velocity with a 65 mph cross wind gust would increase the velocity to 41.09365 meters per second or to 91.9238 mph.
 $0.5 \times 0.1636316 \times (41.09365)^2 = 138.16136542389$ Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.1636316 \times (41.21505)^2 = 138.97889149338$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
 $0.5 \times 0.1636316 \times (41.57712)^2 = 141.43144781218$ Joules

40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.1636316 \times (41.9955)^2 = 144.29214648437$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.1636316 \times (42.17367)^2 = 145.51909066717$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.1636316 \times (42.7034)^2 = 149.19768696348$ Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.1636316 \times (42.93679)^2 = 150.83298549762$ Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.1636316 \times (42.994)^2 = 151.23520019257$ Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.1636316 \times (44.08503)^2 = 159.0081785142$ Joules

55 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.2274 meters per second or to 98.9338 mph.
 $0.5 \times 0.1636316 \times (44.2274)^2 = 160.03685189416$ Joules

70 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 44.2547 meters per second or to 98.994 mph.
 $0.5 \times 0.1636316 \times (44.2547)^2 = 160.23448297682$ Joules

65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
 $0.5 \times 0.1636316 \times (44.36745)^2 = 161.05199851109$ Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.1636316 \times (44.704)^2 = 163.50459046113$ Joules

55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
 $0.5 \times 0.1636316 \times (45.25935)^2 = 167.59220162461$ Joules

70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.1636316 \times (45.86247)^2 = 172.08858462341$ Joules

65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.1636316 \times (46.07983)^2 = 173.72363888698$ Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.1636316 \times (46.5115)^2 = 176.99372636824$ Joules

75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.1636316 \times (47.41575)^2 = 183.94264627441$ Joules

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70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.1636316 \times (47.521)^2 = 184.76015735177$ Joules
65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.1636316 \times (47.83537)^2 = 187.21276448186$ Joules
75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.1636316 \times (49.02176)^2 = 196.61424509677$ Joules
70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.1636316 \times (49.22517)^2 = 198.24928542738$ Joules
80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.1636316 \times (50.5768)^2 = 209.2858531666$ Joules
75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.1636316 \times (50.67549)^2 = 210.10340693887$ Joules
80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.1636316 \times (52.18127)^2 = 222.77500957693$ Joules
85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.1636316 \times (53.73785)^2 = 236.26411010358$ Joules

A 3 inch diameter sphere of ice with a radius of 3.81cm, and a volume of 231.6665944 cm³, and a density of 212.4383 grams would achieve the following energy levels at the following resultant velocities.

40 mph vertical velocity with a 40 mph cross wind gust would increase the fall speed to 25.2884 meters per second or 56 mph.
 $0.5 \times 0.2124383 \times (25.2884)^2 = 67.927483624065$ Joules
40 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 26.91536 meters per second or to 60.207 mph.
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45 mph vertical velocity with a 45 mph cross wind gust would increase the velocity to 28.4494 meters per second or to 63.6395 mph.
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40 mph vertical velocity with a 50 mph cross wind gust would increase the velocity to 28.62453 meters per second or to 64.03125 mph.
 $0.5 \times 0.2124383 \times (28.62453)^2 = 87.032117637154$ Joules
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50 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 40.29564 meters per second or to 90.1387 mph.
 $0.5 \times 0.2124383 \times (40.29564)^2 = 172.47213423387$ Joules

45 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 41.03281 meters per second or to 91.7877 mph.
 $0.5 \times 0.2124383 \times (41.03281)^2 = 178.84027962004$ Joules

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 $0.5 \times 0.2124383 \times (41.09365)^2 = 179.3710114448$ Joules

60 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 41.21505 meters per second or to 92.195 mph.
 $0.5 \times 0.2124383 \times (41.21505)^2 = 180.4323825272$ Joules

55 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 41.57712 meters per second or to 93.005 mph.
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40 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 41.9955 meters per second or to 93.9414 mph.
 $0.5 \times 0.2124383 \times (41.9955)^2 = 187.33043191224$ Joules

50 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 42.17367 meters per second or to 94.3398 mph.
 $0.5 \times 0.2124383 \times (42.17367)^2 = 188.92333900591$ Joules

65 mph vertical velocity with a 70 mph cross wind gust would increase the velocity to 42.7034 meters per second or to 95.524 mph.
 $0.5 \times 0.2124383 \times (42.7034)^2 = 193.69915702379$ Joules

60 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 42.93679 meters per second or to 96.046 mph.
 $0.5 \times 0.2124383 \times (42.93679)^2 = 195.8222190765$ Joules

45 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 42.994 meters per second or to 96.1769 mph.
 $0.5 \times 0.2124383 \times (42.994)^2 = 196.34440309249$ Joules

50 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 44.08503 meters per second or to 98.615 mph.
 $0.5 \times 0.2124383 \times (44.08503)^2 = 206.43584203573$ Joules

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65 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 44.36745 meters per second or to 99.247 mph.
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60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 44.704 meters per second or to 100.000 mph.
 $0.5 \times 0.2124383 \times (44.704)^2 = 212.27340709105$ Joules

55 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 45.25935 meters per second or to 101.242 mph.
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70 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 45.86247 meters per second or to 102.591 mph.
 $0.5 \times 0.2124383 \times (45.86247)^2 = 223.41776506985$ Joules

65 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.07983 meters per second or to 103.0776 mph.
 $0.5 \times 0.2124383 \times (46.07983)^2 = 225.54050999296$ Joules

60 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 46.5115 meters per second or to 104.04 mph.
 $0.5 \times 0.2124383 \times (46.5115)^2 = 229.78597251591$ Joules

75 mph vertical velocity with a 75 mph cross wind gust would increase the velocity to 47.41575 meters per second or to 106.066 mph.
 $0.5 \times 0.2124383 \times (47.41575)^2 = 238.80755961585$ Joules

70 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 47.521 meters per second or to 106.301 mph.
 $0.5 \times 0.2124383 \times (47.521)^2 = 239.8689112344$ Joules

65 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 47.83537 meters per second or to 107.0096 mph.
 $0.5 \times 0.2124383 \times (47.83537)^2 = 243.05306202975$ Joules

75 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 49.02176 meters per second or to 109.658 mph.
 $0.5 \times 0.2124383 \times (49.02176)^2 = 255.2587396575$ Joules

70 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 49.22517 meters per second or to 110.113 mph.
 $0.5 \times 0.2124383 \times (49.22517)^2 = 257.38146649184$ Joules

80 mph vertical velocity with an 80 mph cross wind gust would increase the velocity to 50.5768 meters per second or to 113.137 mph.
 $0.5 \times 0.2124383 \times (50.5768)^2 = 271.70993449626$ Joules

75 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 50.67549 meters per second or to 113.67549 mph.
 $0.5 \times 0.2124383 \times (50.67549)^2 = 272.77133875304$ Joules

80 mph vertical velocity with and 85 mph cross wind gust would increase the velocity to 52.18127 meters per second or to 116.726 mph.
 $0.5 \times 0.2124383 \times (52.18127)^2 = 289.22252374851$ Joules

85 mph vertical velocity with an 85 mph cross wind gust would increase the velocity to 53.73785 meters per second or to 120.208 mph.
 $0.5 \times 0.2124383 \times (53.73785)^2 = 306.73504323992$ Joules

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

Additional References:

The book entitled "Gravity" written by George Gamow; pages 24-25 "If you had a long glass cylinder evacuated of air, you would see that a coin, an un-crumpled piece of paper, and a feather would fall inside the cylinder at exactly the same speed." –Describing the motion of falling objects inside a vacuum which "free fall" under the sole force of gravity.

The book entitled "Gravity" written by George Gamow; Page 33, "Using the data obtained by experiments with the inclined planes, one can find that the acceleration of free fall is 386.2 inches per second squared (you probably are familiar with the equivalent expression "32.2 feet per second per second") or in the metric system, 981 cm/sec²." –Describing the results of obtained by Galileo's experiments with rolling balls made of different materials that were sent down inclined planes so that the acceleration of the rolling balls could be lengthened by a factor of the slope to create a ratio that was slow enough to allow the acceleration of gravity to become measurable.

The book entitled "Elementary Physics, Atoms, Waves, Particles" written by George A Williams; Page 32 the formula for calculating the length of a free falling object: Distance = (velocity)²/2xAcceleration

An example of this formula: What is the distance it will take for a "free-falling object" to reach 50 miles per hour? Distance = (73.333333²)/(2x32.1850393) = a distance of 83.544 feet

The book entitled eighth edition Volume 1 "Physics for Scientists and Engineers"; Page 156 "In the limiting case for a falling object, when the magnitude of the resistive force equals the objects weight, the object reaches its terminal speed." – Describing the key difference between free falling objects which continue accelerating for as long as the free falling objects are permitted to remain fallings, where as non-free falling objects that fall through resistive forces such as air will reach a maximum limit of acceleration when the weight of the falling object becomes equal with its surrounding resistive forces.

The book entitled "Fundamentals of Physics 8th Edition" Written by Jearl Walker

The book entitled "Encyclopedia of Physical Science" Volume 1 & Volume 2 written by Joe Rosen, P.H.D. and Lisa Quinn Gothard

The book entitled "A first course of physical laboratory practice" written by Arthur Mason Worthington

The workbook entitled "Haag certified roof inspector program residential edition course workbook

The Publication entitled "Hail: Sizing it up!" by Vickie Crenshaw and Jim D. Koontz

The Publication entitled "What are the effects of hail on residential roofing products?" by Jim D. Koontz

Comparable Research Examination:

The only other resultant velocity hail ice ball impact laboratory research that I could find related to ice balls being propelled at resultant velocities is the publication written by Vickie Crenshaw and Jim D. Koontz, entitled "Hail Sizing it up!"

In the "Hail sizing it up publication," these researcher conducted testing with ice 1" inch, 1.5", 2", 2.5", & 3" ice spheres.

These researchers found that 1" diameter ice spheres, with a mass of 7.85 grams, that were propelled at a 103 feet per second resultant velocity produced 2.57 pound feet of energy, and that,

1.5 inch diameter ice spheres, with a mass of 26.50 grams, that were propelled at a 127 feet per second resultant velocity produced 14.72 pound feet of energy and that,

2 inch diameter ice spheres, with a mass of 62.81 grams, that were propelled at a 148 feet per second resultant velocity produced 47.49 pound feet of energy. and that,

2.5 inch diameter ice spheres, with a mass of 122.67 grams, that were propelled at a 165 feet per second resultant velocity produced 115.17 pound feet of energy and that,

3 inch diameter ice spheres, with a mass of 211.98 grams, that were propelled at a 183 feet per second resultant velocity produced 245.70 pound feet of energy.

So to compare the accuracy of my own calculations with their published calculations I converted their velocities from feet per second to meters per second, and their energy levels from foot pounds to joules, and found that my calculations were within 1% of their calculations. Our masses are very slightly different because I assumed that the medium density of solid ice was .917kg, and they used .91, but its a very small difference.

There is one error in their calculations which is that according to Vickie Crenshaw and Jim D. Koontz a 1 inch diameter ice sphere traveling at a 73 feet per second (31.394m/s/s) velocity would produce 2.57 foot pounds. When in reality it would create 2.853 foot pounds, or 3.868 joules $0.5 \times 0.00785 \times (31.394)^2 = 3.8684142013$ joules.

Otherwise,

1" hail stone: They tested with 103 feet per second which equals 31.394 meters per second, compared to my closest comparable resultant velocity for a 1" ice sphere which is 30.40201 meters per second which resulted in $(.5 \times 0.007868) \times (30.40201)^2 = 3.63613$ Joules or 2.681871858 foot pounds, (again my velocity was slower and they erred by publishing 2.57 foot pounds) when it is really 2.853 foot pounds or 3.868 joules at their faster speed.

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

1.5 inch Ice Sphere: They tested with 127 feet per second which equals 38.709 meters per second, and said that it would produce 14.72 foot pounds or 19.957 joules, compared to my closest resultant velocity for a 1.5 inch ice sphere which is 38.45583 meters per second, and I got $0.5 \times 0.0265547 \times (38.45583)^2 = 19.635220479151$ Joules or 14.482195419 foot pounds at a slightly slower velocity.

2 inch Ice Sphere: They tested with 148 feet per second which equals 45.110 meters per second, and said that it would produce 47.49 foot pounds or 64.386 joules, compared to my closest resultant velocity for a 2 inch hail stone which is 45.25935 meters per second, and I got $0.5 \times 0.06294468 \times (45.25935)^2 = 64.46821702994$ Joules or 47.549316714 foot pounds with a slightly faster velocity.

2.5 inch Ice Sphere: They tested with 165 feet per second which equals 50.292 meters per second, and they said that it would produce 115.17 foot pounds or 156.147 joules, compared to my closest resultant velocity for a 2.5 inch ice sphere which was 50.5768 meters per second, and I got $0.5 \times 0.12293884 \times (50.5768)^2 = 157.23955691345$ Joules or 115.973945552 foot pounds with a slightly faster velocity.

My conclusion is that that my resultant velocity and energy level calculations compared very well with the resultant velocity and energy level calculations that were published by Vickie Crenshaw and Jim D Kootz as display on Table 4 of their "Hail Sizing it up!" publication.

Below are the original calculations.

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

1.) Radius of 1.273 cm, a volume of 8.58024664 cm³, and a density of 7.868 grams
 $(.5)(.007868)(497.29) = 1.956 \text{ Joules}$ ^{11.43 ft/lb}
 or at 18.16 m/s = 1.36 Joules 41.607 mph 58 ft/lb 1.003

1.25) Radius of ~~1.905~~ 1.5875 cm, a volume of 16.75829423 cm³, and a density of 15.367355 grams
 $(.5 \times .0153673)(45^2) = 4.08022 \text{ Joules}$ 3.5419 ft/lb 0.153673 kg
 or at 26.57 m/s = 5.424 Joules 59.935397 mph 4.000 ft/lb

1.5) Radius of ~~1.905~~ 1.905 cm, a volume of 28.9583324 cm³ and a density of 26.5547 grams
 $(.5 \times .0265547)(27.4^2) = 9.968103 \text{ Joules}$ 7.35209 ft/lb 0.0265547 kg
 or at 28.59 m/s = 10.85 63.954008 mph 8.00254 ft/lb

1.75) Radius of 2.2225 cm, a volume of 45.98475936, density of 42.16802433
 $(.5 \times .04216802)(29.6^2) = 18.4729 \text{ Joules}$ 13.6249 ft/lb 0.04216802 kg
 $(29.99)^2 = 18.96 \text{ Joules}$ 67.08795 mph 13.98 ft/lb

2) Radius of 2.54 cm, a volume of 68.64197316, and a density of 62.94468 grams
 ~~$(.5 \times .06294468)(32^2) = 32.22747616 \text{ Joules}$~~ 23.769 ft/lb 0.06294468 kg
 $(.5 \times .06294468)(32^2) = 32.22747616 \text{ Joules}$ 68.85289 mph 21.986 ft/lb
 or (30.78) = 29.81 Joules

2.25) Radius of 2.8575 cm, a volume of 97.73437195, density of 89.622419
 $(.5 \times .08962241)(34.0)^2 = 51.80175 \text{ Joules}$ 38.20701 ft/lb 0.08962241 kg
 $(32.04)^2 = 46.01 \text{ Joules}$ 71.6803 mph 33.935 ft/lb

2.5) Radius of 3.175 cm, a volume of 134.00663538, and a density of 122.93889 grams
 $(.5 \times .12293889)(35.6^2) = 78.34216105 \text{ Joules}$ 57.782 ft/lb 0.12293889 kg
 $(34.2)^2 = 71.939 \text{ Joules}$ 76.525 mph 53.0554 ft/lb

2.75) Radius of 3.4925 cm, a volume of 178.44231696, density of 163.631604 grams
 $(.5 \times .163631604)(37.6^2) = 115.16679 \text{ Joules}$ 85.3122 ft/lb 0.163631604 kg
 or (36.6)^2 = 109.8368 Joules 81.961 mph 81.00654 ft/lb

3) Radius of 3.81 cm, a volume of ~~221.4383~~ 231.6665944, and a density of 212.4383 grams
 $(.5 \times .2124383)(39.6^2) = 166.5686226 \text{ Joules}$ 122.88 ft/lb 0.2124383 kg
 $(39.6)^2 = 162.388 \text{ Joules}$ 87.4642 mph 119.77 ft/lb
 67.725
 618153) 633

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

1)

50 mph 1.4426 Ft/lbs (44.6 m/s) 7.825355 Joules

$$(.5)(.007868)(22.3^2) = 1.956 \text{ Joules}$$

$$497.29 + 497.29 = 994.58 = (31.536^2) \text{ m/s}$$

$$(.5)(.007868)(25.0^2) = 2.45875 \text{ Joules}$$

$$(625)$$

$$= 3912438 \text{ Joules}$$

61 mph

$$(.5)(.007868)(27.4^2) = 2.953489 \text{ Joules}$$

$$(750.76)$$

66 mph

$$(.5)(.007868)(29.6^2) = 3.44681344 \text{ Joules}$$

$$(876.16)$$

72 mph

$$(.5)(.007868)(32.0^2) = 4.028416 \text{ Joules}$$

$$1024$$

76 mph

$$(.5)(.007868)(34.0^2) = 4.547704 \text{ Joules}$$

$$1156$$

80 mph

$$(.5)(.007868)(35.7^2) = 5.0613843 \text{ Joules}$$

$$1274.49$$

84 mph

$$(.5)(.007868)(37.1^2) = 5.56173184 \text{ Joules}$$

$$1413.76$$

88 mph

$$(.5)(.007868)(39.2^2) = 6.16914144 \text{ Joules}$$

$$1568.16$$

12/12	45°	5/12	22.62
11/12	42.51	4/12	18.143
10/12	39.81	3/12	14.04
9/12	36.87	2/12	9.81
8/12	33.69		
7/12	30.26		
6/12	26.57 26.57		

$V_f = V_i + a \cdot t$

Handwritten calculations for ice ball impact testing, showing various formulas and results.

Top Section:

- 1) 7.85 Gm
- 1.5) 26.50 Gm
- 5.082) 62.81 Gm
- 6.325) 122.67 Gm
- 7.623) 211.98 Gm

Middle Section:

700 Kg car at 20 m/s

$$KE = \frac{1}{2} \times 700 \text{ Kg} \times 20^2 = 140,000 \text{ Joules}$$

Bottom Section:

Volume

- 1) 8.58029664
- 1.5) 28.95833232
- 2) 68.64197316
- 2.5) 134.066353
- 3) 231.6665944

Results:

- 1) 7.868 gram
- 1.5) 26.5547 grams
- 2) 62.94468 gram
- 2.5) 122.93884 grams
- 3) 212.4383 grams

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

Kootz			Haug		
Ft/sec	lb/Ft	Ft to m/sec	m/sec		
1 73	2.57	22.25 22.25	1 22.3	David Grinstead 11200 Myrtle Ave garden 661-794-3169 David Grinstead agent Myrtle Ave	
1.5 90	14.72	27.4	1.5 27.4		
2 105	47.49	32.14	2 22.6 ³²		
2.5 117	115.17	35.6	2.5 35.7		
3 130	245.7	39.6	3 39.6		

Result v m/s			Kootz and Ft Haug		
1	31.94		1	2.57	<1
1.5	38.7		1.5	14.72	14.72 8
2	45.1		2	47.49	22
2.5	50.29		2.5	115.17	53
3	55.77		3	245.70	120

75.65 10 Gram
= 0.02204616

$(9.8)^2 \times 25$
2401 meters - 7877 feet

$(9.8)^2 \times 27.4 = 2631 \text{ meters } 8631$

105.4 Altitude or 32,1259 meters high

32,152 feet per second $(32,1259)^2 \times (9.8)^2$
for 3.2781 seconds = 99120
 $\sqrt{1.81055240189}$

Ice = .936m/cu. cm = 8,55857 Gram
.00093 kg = .018868 lbs

1-inch = .0254 = 1033.76 .80 cross sect

$9.8 \text{ m/s} = 32,1522 \text{ feet per second}$.07 Gram
 $\times 32,1259$ 105.4 ft
= 3388

56 mph = 82,133,373,334 feet per second
 $\sqrt{82,133,373,373,334}^2 = 6245$
= 9,062,74424

57.3 110.34 $\sqrt{10,504}$ = 11,77725
3,431,802,489 $\sqrt{110.34}$
32,1522 = 1033.76 = 12174,84996 $\sqrt{110.34}$
3,431,808 = 11,77725 5,965,232 10
110.33970 $\sqrt{3,02776}$
84,039,999 $\sqrt{9,167,333}$

$84,039,999 / 32,1522 = 2,613,818$
 $2,613,818^2$
= 6,832,04466
=

Sum of time squared
Plus Rate squared
+ Square of the Square of the final

$\frac{1}{2} \text{ of } 32,1522^2$ Speed
 $32,1522 = 16,0761$ of
 $\times 16,0761^2 = 258,140,9921$ 84,039,999
 $16,07612 \times 3,431,808^2$ $\sqrt{9,167,333}$
 $\sqrt{3,02776}$

~~of time 11,7725~~
~~Speed 1033,76~~
2 of feet per second 1033,76
2 of time (s) 11,7725 $\times = 12174,84996$
 $\sqrt{12174,84996} = 110.34$
 $t = \sqrt{\frac{2H}{g}}$
 $= \sqrt{\frac{2(110.34)}{9.8 \text{ m/s}^2}} \sqrt{10,504,92369}$
 $=$
 $110.34 \times 2 = 220.68 / 32,1522^2$
1033,76391484

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

$$\begin{aligned}
 & 1 \frac{1}{4} \text{ 57.3 mph } \frac{84,039999}{32.152} = 110 \text{ ft} \\
 & 1 \frac{1}{2} \text{ 63.1 mph } \frac{92,546666}{32.1522} = 2,878392^2 \\
 & \quad = 8,285146 \\
 & \quad 16.076212 \times 8,285146^2 = 133 \text{ ft} \\
 & 1 \frac{3}{4} \text{ 69.4 mph } 101,786666 \text{ ft/s}^2 \\
 & \quad \frac{16.076212}{32.1522} = 6,3315081 \text{ ft/s}^2 \\
 & \quad \frac{16.076212}{32.1522} = 40,8879949 \\
 & \quad = 3,16577611 \\
 & \quad 3,16577611^2 = 10.0222138 \\
 & \quad 16,076212 \times 10.0222138 \\
 & \quad = 161 \text{ ft} \\
 \hline
 & VT = (0.222 * g * (dh - da) * (r ** 2)) / N \\
 & G = 9.8 \text{ m/s}^2 \\
 & dh = \text{density of the hail } (0.917 \text{ g/cm}^3 \text{ for ice}) \\
 & da = \text{density of air (Approximate } 0.0012932 \text{ g/cm}^3) \\
 & R = \text{Radius (half the diameter of hail stone in cm)} \\
 & N = \text{dynamic viscosity of air near the Earth's surface approx } 0.0018 \text{ g/cm}^2
 \end{aligned}$$

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

1 = 50 mph	$\frac{23.33333}{32,1850393}$ $\frac{2,27849133}{\text{fall time } 16.09251965}$	83.544 ft
1.25 = 56 mph	$\frac{82.133333}{32,1850393 = \text{SQ} = 6.5122448 \times 16.09251965}$ $\frac{2.551910301}{\text{fall time}}$	104.798 ft
1.5 = 61 mph	$\frac{89.466666}{\text{speed} = \text{SQ} = 7.72706251 \times 16.0925}$ $\frac{2.779759434}{\text{fall time}}$	124 ft
1.75 = 66 mph	$\frac{96.8}{\text{speed} = \text{SQ} = 9.04570930 \times 16.0925}$ $\frac{3.07766856}{\text{fall time}}$	145.56 ft
2 = 72 mph	$\frac{105.6}{\text{speed} = \text{SQ} = 10.76514165 \times 16.0925}$ $\frac{3.28102753}{\text{fall time}}$	173.238 ft
2.25 = 76 mph	$\frac{111.46666}{\text{speed} = \text{SQ} = 11.99449424 \times 16.0925}$ $\frac{3.46330683}{\text{fall time}}$	193.021 ft
2.5 = 78 80 mph	$\frac{117.3333}{\text{speed} = \text{SQ} = 13.29029833 \times 16.0925}$ $\frac{3.64558614}{\text{fall time}}$	213.874 ft
2.75 = 84 mph	$\frac{123.2}{\text{speed} = \text{SQ} = 14.65255391 \times 16.0925}$ $\frac{3.82786545}{\text{fall time}}$	235.796 ft
3 = 88 mph	$\frac{129.0666}{\text{speed} = \text{SQ} = 16.08126098 \times 16.0925}$ $\frac{4.01014475}{\text{fall time}}$	258.788 ft

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

A handwritten table on lined paper showing the relationship between wind speed in mph and resultant velocity. The table lists wind speeds from 30 to 90 mph and their corresponding resultant velocities. The values are written in blue ink.

30 mph	13.4112
45 miles mph	20.11679
50 mph	22.352
60 mph	26.8224
65 mph	29.0576
70 mph	31.2928
75 mph	33.528
80 mph	35.7632
85 mph	37.9984
85	37.99
90	40.2335

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

A handwritten table on lined paper showing the conversion of various speeds in miles per hour (mph) to a decimal value. The table lists speeds from 30 mph to 90 mph in increments of 5 mph. The values are written in blue ink. The 45 mph entry has a correction from 'miles' to 'mph'. The 80 mph entry has a correction from '35.7632' to '35.7632'.

30 mph	13.4112
45 miles mph	20.11679
50 mph	22.352
60 mph	26.8224
65 mph	29.0576
70 mph	31.2928
75 mph	33.528
80 mph	35.7632
35.7632	
85	37.99
90	40.2335

14) 7.75×16.75829827
 $= 386.38189 \text{ grams}$ or 129.87681 gms
or 386.38189 kg or 129.87681
Dropped 12 ft High = 18.90 mph
or 8.449056 m/s
 $(.5 \times 386.38189) \times (8.449056^2)$
 $(.5 \times 129.87681) \times (8.449056^2) = 4,163.573 \text{ Joules}$

15) $7.75 \times 28.9583324 = 224,427.08 \text{ grams}$
 $= 224.42708 \text{ kg}$
Dropped 14 ft = 20.91 mph
 $= 9.124086 \text{ m/s}$
 $(.5 \times 224.42708) \times (9.124086^2) = 9,341.66$

175) $7.75 \times 45.98475936 = 356.38189 \text{ kg}$
or 356.38189 kg
Dropp 17 ft High @ 22.49 mph
or 10.0539296 m/s
 $(.5 \times 356.38189) \times (10.0539296^2) = 18,011.81 \text{ Joules}$

007868 kg, 1.27 CS, Drag Coef. 5, density of .917
 $V = \text{acc/gram} \times \text{Diameter}^2 \times (\rho \text{ Descent Particle} - \text{Density of medium})$
18 X Viscosity of medium
Iron $7.75 \times 68,641.9734$
Volume of 2" sphere
 $= 531,975.29 \text{ grams}$
 $.53197529 \text{ kg}$
Dropped 20 ft High = 24.39 mph
 $24.39 \text{ mph} = 10.9037056 \text{ m/s}$
 $= 31,621.16 \text{ Joules}$
or 23,322.5707 foot-pounds

58.5" m
.5" @ 16 ft = .14913997
.75" @ 16 ft = .67908974
1.0" @ 16 ft = 2.074401461 Joules
1.25" @ 16 ft = 5.043642765
1.5" @ 16 ft = 10.534705458
1.75" @ 16 ft = 14.941114
2.0" @ 16 ft = 26.80167
2.25" @ 16 ft = 36.335921015
2.5" @ 16 ft = 43.65116
2.75" @ 16 ft = 59,222.12798

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

☆ 1,235 5) 3/4 1659 6) 3/4 2300									
FM473									
O and 10%+									
1,25	.03381lb (15.3) gm								
1,5	.05891lb (26.5) gm								
1,75	.09281lb (42.1) gm								
2	.13851lb (62.9) gm								
Selkirk Type D Saw Wood Household MFG, TYPE B — 95 KSI A129-4C Stainless Steel									
.02 Carbon .50 Manganese .03 Phosphorus .01 Sulfur .55 Silicon 2.9 Chromium .3 Nickel									
4 molybdenum .02 Nitrogen .16 Nickelium Rest Iron									
34.83 65.17 Iron									
		MPH	Ft/s	MPH	Ft/s	MPH	Ft/s	MPH	Ft/s
		10	14.66	31	45.466	52	76.266	73	107.666
		11	16.133	32	46.933	53	77.73	74	109.133
		12	17.6	33	48.4	54	79.2	75	110
		13	19.066	34	49.866	55	80.666	76	111.466
		14	20.533	35	51.333	56	82.133	77	112.933
		15	22	36	52.8	57	83.6	78	114.4
		16	23.466	37	54.266	58	85.066	79	115.866
		17	24.933	38	55.733	59	86.533	80	117.333
		18	26.4	39	57.2	60	88	81	118.8
		19	27.866	40	58.666	61	89.466	82	120.266
		20	29.333	41	60.133	62	90.933	83	121.733
		21	30.8	42	61.6	63	92.4	84	123.2
		22	32.266	43	63.066	64	93.866	85	124.666
		23	33.733	44	64.533	65	95.333	86	126.133
		24	35.2	45	66	66	96.8	87	127.6
		25	36.666	46	67.466	67	98.266	88	129.066
		26	38.133	47	68.933	68	99.733	89	130.533
		27	39.6	48	70.4	69	101.2	90	132
		28	41.066	49	71.866	70	102.666	91	133.466
		29	42.533	50	73.333	71	104.133	92	134.933
		30	44	51	74.8	72	105.6	93	136.4

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

Yb Truggle 31105 S. Fallbrook Ct.				Yb Truggle © SBC Global.net				MPH to meters per second (10 MPH to 72 MPH)			
mph	Ft/s	mph	Ft/s	mph	m/s	mph	m/s	mph	m/s	mph	m/s
94	137.866	115	167.666			10	4.4704	31	13.85824	52	23.24608
95	139.273	116	170.133			11	4.91744	32	14.30528	53	23.69312
96	140.8	117	171.6			12	5.36448	33	14.75232	54	24.14016
97	142.266	118	173.066			13	5.81152	34	15.19936	55	24.5872
98	143.733	119	174.533			14	6.25856	35	15.6464	56	25.03424
99	145.2	120	176			15	6.7056	36	16.09344	57	25.48128
100	146.666	121	177.466			16	7.15264	37	16.54048	58	25.92832
101	148.133	122	178.933			17	7.59968	38	16.98752	59	26.37536
102	149.6	123	180.4			18	8.04672	39	17.43456	60	26.8224
103	151.066	124	181.866			19	8.49376	40	17.8816	61	27.26944
104	152.533	125	183.333			20	8.9408	41	18.32864	62	27.71648
105	154	126	184.8			21	9.38784	42	18.77568	63	28.16352
106	155.466	127	186.266			22	9.83488	43	19.22272	64	28.61056
107	156.933	128	187.733			23	10.28192	44	19.66976	65	29.0576
108	158.4	129	189.2			24	10.72896	45	20.1168	66	29.50464
109	159.866	130	190.666			25	11.176	46	20.56384	67	29.95168
110	161.333	131	192.133			26	11.62304	47	21.01088	68	30.39872
111	162.8	132	193.6			27	12.07008	48	21.45792	69	30.84576
112	164.266	133	195.066			28	12.51712	49	21.90496	70	31.2928
113	165.733	134	196.533			29	12.96416	50	22.352	71	31.73984
114	167.2	135	199			30	13.4112	51	22.79904	72	32.18688

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

mph to m/s per second
(73 mph to 135 mph)

mph	m/s	mph	m/s	mph	m/s
73	32.63392	94	42.02176	115	51.4096
74	33.0896	95	42.47744	116	51.86528
75	33.54528	96	42.93312	117	52.32096
76	33.97504	97	43.3888	118	52.77664
77	34.4208	98	43.84448	119	53.23232
78	34.86656	99	44.256	120	53.6448
79	35.31232	100	44.704	121	54.10048
80	35.75808	101	45.15968	122	54.55616
81	36.20384	102	45.61536	123	55.01184
82	36.6496	103	46.07104	124	55.46752
83	37.09536	104	46.52672	125	55.9232
84	37.54112	105	46.9824	126	56.37888
85	37.98688	106	47.43808	127	56.83456
86	38.43264	107	47.89376	128	57.29024
87	38.8784	108	48.34944	129	57.74592
88	39.32416	109	48.80512	130	58.2016
89	39.76992	110	49.2608	131	58.65728
90	40.21568	111	49.71648	132	59.11296
91	40.66144	112	50.17216	133	59.56864
92	41.1072	113	50.62784	134	60.02432
93	41.55296	114	51.08352	135	60.48

James O'Donnell / Down Falls
810-223-5411 60 E 417

mph	m/s	Resultant Circle
40	40	= 25.12004 m/s or 56 mph
40	41	= 25.6043 m/s or 57.275 mph
40	42	= 26.0886 m/s or 57.999 mph
40	43	
40	45	= 26.91536 m/s or 60.2079 mph
40	50	= 28.62453 m/s or 64.03125 mph
40	55	= 30.40201 m/s or 68.161735 mph
40	60	= 32.12361 m/s or 72.1111 mph
40	65	= 34.1185 m/s or 76.32169 mph
40	70	= 36.04152 m/s or 80.62258 mph
40	75	= 37.99894 m/s or 84.999 mph
40	80	= 39.98447 m/s or 89.4427 mph
40	85	= 41.9954 m/s or 93.94148 mph

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

mph			mph		
45	40 mph gust	$\approx 26,91576 \text{ m/s}$ or $60,267 \text{ mph}$	60	40	$\approx 32,23651 \text{ m/s}$ or $72,111 \text{ mph}$
45	45	$\approx 28,4444 \text{ m/s}$ or $63,6595 \text{ mph}$	60	45	$\approx 33,52799 \text{ m/s}$ or $74,999 \text{ mph}$
45	50	$\approx 30,07153 \text{ m/s}$ or $67,2680 \text{ mph}$	60	50	
45	55	$\approx 31,76015 \text{ m/s}$ or $71,06332 \text{ mph}$	60	55	$\approx 34,4201 \text{ m/s}$ or $76,80735 \text{ mph}$
45	60	$\approx 33,52799 \text{ m/s}$ or $74,999 \text{ mph}$	60	60	$\approx 36,38642 \text{ m/s}$ or $81,394 \text{ mph}$
45	65	$\approx 35,34111 \text{ m/s}$ or $79,05693 \text{ mph}$	60	65	$\approx 38,16408 \text{ m/s}$ or $85,06408 \text{ mph}$
45	70	$\approx 37,20114 \text{ m/s}$ or $83,2165 \text{ mph}$	60	70	$\approx 39,79661 \text{ m/s}$ or $89,022 \text{ mph}$
45	75	$\approx 39,10003 \text{ m/s}$ or $87,464276 \text{ mph}$	60	75	$\approx 41,57712 \text{ m/s}$ or $93,065 \text{ mph}$
45	80	$\approx 41,03281 \text{ m/s}$ or $91,7877 \text{ mph}$	60	80	$\approx 44,2274 \text{ m/s}$ or $98,9338 \text{ mph}$
45	85	$\approx 42,99493 \text{ m/s}$ or $96,1769 \text{ mph}$	60	85	$\approx 45,25935 \text{ m/s}$ or $101,242 \text{ mph}$
50	40	$\approx 28,162453 \text{ m/s}$ or $64,183126 \text{ mph}$	60	40	$\approx 32,23651 \text{ m/s}$ or $72,111 \text{ mph}$
50	45	$\approx 30,07153 \text{ m/s}$ or $67,2680 \text{ mph}$	60	45	$\approx 33,52799 \text{ m/s}$ or $74,999 \text{ mph}$
50	50	$\approx 31,6105 \text{ m/s}$ or $70,71067 \text{ mph}$	60	50	$\approx 34,91494 \text{ m/s}$ or $78,102496 \text{ mph}$
50	55	$\approx 33,22864 \text{ m/s}$ or $74,3303 \text{ mph}$	60	55	$\approx 36,38642 \text{ m/s}$ or $81,394 \text{ mph}$
50	60	$\approx 34,91494 \text{ m/s}$ or $78,102496 \text{ mph}$	60	60	$\approx 37,9362 \text{ m/s}$ or $84,8528 \text{ mph}$
50	65	$\approx 36,6601 \text{ m/s}$ or $82,066166 \text{ mph}$	60	65	$\approx 39,54472 \text{ m/s}$ or $88,4959 \text{ mph}$
50	70	$\approx 38,45583 \text{ m/s}$ or $86,10232 \text{ mph}$	60	70	$\approx 41,21505 \text{ m/s}$ or $92,195 \text{ mph}$
50	75	$\approx 40,29564 \text{ m/s}$ or $90,13877 \text{ mph}$	60	75	$\approx 42,93679 \text{ m/s}$ or $96,046 \text{ mph}$
50	80	$\approx 42,17367 \text{ m/s}$ or $94,3798 \text{ mph}$	60	80	$\approx 44,702 \text{ m/s}$ or $100,002 \text{ mph}$
50	85	$\approx 44,08503 \text{ m/s}$ or $98,6150 \text{ mph}$	60	85	$\approx 46,5115 \text{ m/s}$ or $104,04 \text{ mph}$

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

65	40	= 34.11175 or 76.32169 mph
65	45	= 35.34161 or 79.05693 mph
65	50	= 36.66001 m/s or 82.006106 mph
65	55	= 38.06405 m/s or 85.06408 mph
65	60	= 39.54472 m/s or 88.459 mph
65	65	= 41.05313 m/s or 91.92387 mph
65	70	= 42.70344 m/s or 95.524 mph
65	75	= 44.36745 m/s or 99.247 mph
65	80	= 46.07983 m/s or 103.0776 mph
65	85	= 47.83537 m/s or 107.00967 mph
70	40	= 36.04152 or 80.62258 mph
70	45	= 37.20114 m/s or 83.2165 mph
70	50	= 38.45683 m/s or 86.0232
70	55	= 39.7966 m/s or 89.02 mph
70	60	= 41.21505 m/s or 92.195 mph
70	65	= 42.70344 m/s or 95.524 mph
70	70	= 44.2547 or 98.994 mph
70	75	= 45.86247 or 102.591 mph
70	80	= 47.521 m/s or 106.301 mph
70	85	= 49.22517 or 110.113 mph
75	40	= 37.9984 m/s or 84.999 mph
75	45	= 39.1603 or 87.464276 mph
75	50	= 40.29564 or 90.13877 mph
75	55	= 41.57712 or 93.009 mph
75	60	= 42.93679 m/s or 96.046 mph
75	65	= 44.36745 m/s or 99.247 mph
75	70	= 45.86247 or 102.591 mph
75	75	= 47.41575 or 106.066 mph
75	80	= 47.521 m/s 49.62176 or 109.658 mph
75	85	= 50.67549 or 113.167549 mph
80	40	= 39.98447 m/s or 89.4427 mph
80	45	= 41.03281 m/s or 91.7877 mph
80	50	= 42.17367 or 94.3398 mph
80	55	= 44.2274 or 98.9385 mph
80	60	= 46.704 m/s or 100.000 mph
80	65	= 46.704 46.67983 m/s or 103.0776 mph
80	70	= 47.521 m/s or 106.301 mph
80	75	= 49.02176 m/s or 109.658 mph
80	80	= 50.5768 m/s or 113.137 mph
80	85	= 52.18127 m/s or 116.726 mph

A closer look at Ice ball impact testing, and the true resultant velocity of real life hail stones.

85	40 =	41,995 m/s or 93,94142 mph
85	45 =	42,99493 or 96,1769 mph
85	50 =	44,08503 or 98,615
85	55 =	45,25935 or 101,242 mph
85	60 =	46,5115 or 104,64
85	65 =	47,83537 or 107,00967 mph
85	70 =	49,22517 or 110,113 mph
85	75 =	50,67549 or 113,067549 mph
85	80 =	52,18127 m/s or 116,726 mph
85	85 =	53,73784 53,73785 m/s or 120,208 mph